# Sieve of Eratosthenes (1–100)

**Mathematical description.** Fix $n = 100$. Let $S = \{2, 3, \dots, n\}$. For each integer $p$ defined as the smallest element of $S$ not yet processed, declare $p$ *prime* and remove from $S$ every integer of the form $kp$ with $k \geq 2$. Continue this procedure while $p^2 \leq n$. The elements of $S$ that remain after this process are precisely the prime numbers $\leq n$.

**Algorithm (short).**

1. Initialize the list $2, 3, \dots, n$.
2. For each current smallest unprocessed $p$ (starting at $p = 2$), remove all multiples $2p, 3p, 4p, \dots$ up to $n$.
3. Stop once $p^2 > n$ (i.e. $p > \lfloor\sqrt{n}\rfloor$). Remaining numbers are primes.

**Note.** 1 is neither prime nor composite.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | **2** | **3** | 4 | **5** | 6 | **7** | 8 | 9 | 10 |
| **11** | 12 | **13** | 14 | 15 | 16 | **17** | 18 | **19** | 20 |
| 21 | 22 | **23** | 24 | 25 | 26 | 27 | 28 | **29** | 30 |
| **31** | 32 | 33 | 34 | 35 | 36 | **37** | 38 | 39 | 40 |
| **41** | 42 | **43** | 44 | 45 | 46 | **47** | 48 | 49 | 50 |
| 51 | 52 | **53** | 54 | 55 | 56 | 57 | 58 | **59** | 60 |
| **61** | 62 | 63 | 64 | 65 | 66 | **67** | 68 | 69 | 70 |
| **71** | 72 | **73** | 74 | 75 | 76 | 77 | 78 | **79** | 80 |
| 81 | 82 | **83** | 84 | 85 | 86 | 87 | 88 | **89** | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | **97** | 98 | 99 | 100 |

**Primes** The primes $\leq 100$ are:

**2**, **3**, **5**, **7**, **11**, **13**, **17**, **19**, **23**, **29**,**31**, **37**, **41**, **43**, **47**, **53**, **59**, **61**, **67**, **71**,**73**, **79**, **83**, **89**, **97**.

## 0.1 L1: 1.1 Divisibility (syllabus, expectations)

## 0.2 L2: 1.2 Prime numbers (open conjectures, breakthroughs)

### 0.3 L3: 1.3 GCD, 1.4 Euclidean Algorithm

## 0.4 L4: 1.5 Fundamental Theorem of Arithmetic

## 0.5 L5: 1.5 Dirichlet's Theorem, 2.1 Congruences

# Lecture 5: Dirichlet's Theorem and applications of the FTA

**§ Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions)** Let $a, b \in \mathbb{Z}$ with $b > 0$ and $\gcd(a, b) = 1$. Then the arithmetic progression

$$a,\ a+b,\ a+2b,\ a+3b,\ \ldots$$

contains infinitely many prime numbers.

**§ Remark (on Theorem Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions))** Setting $a = 1$ and $b = 1$ recovers the elementary statement that there are infinitely many primes. (We remark that the full proof of Dirichlet's theorem requires analytic tools and is beyond the scope of this course.)

**§ Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4)** There are infinitely many primes of the form $4n + 3$, where $n$ is a nonnegative integer.

**§ Lemma 1.23 (Product of numbers congruent to 1 modulo 4)** Let $a, b \in \mathbb{Z}$. If $a \equiv 1 \pmod 4$ and $b \equiv 1 \pmod 4$, then $ab \equiv 1 \pmod 4$.

*Proof.* Write $a = 4n_1 + 1$ and $b = 4n_2 + 1$ with $n_1, n_2 \in \mathbb{Z}$. Then

$$ab = (4n_1 + 1)(4n_2 + 1) = 16n_1n_2 + 4n_1 + 4n_2 + 1 = 4(4n_1n_2 + n_1 + n_2) + 1,$$

so $ab \equiv 1 \pmod 4$. This proves the lemma. □

*Proof of Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4).* Suppose, for contradiction, that there are only finitely many primes congruent to 3 modulo 4. List them as

$$P_1 = 3,\ P_2,\ P_3,\ \ldots,\ P_r,$$

all distinct and each $P_i \equiv 3 \pmod 4$.

Consider the integer

$$N := 4P_1P_2 \cdots P_r - 1.$$

Note that $N \equiv -1 \equiv 3 \pmod 4$.

Let $q$ be any prime divisor of $N$. We make two observations:

1. For each $i = 1, \ldots, r$ we have $N \equiv -1 \pmod{P_i}$, so $P_i \nmid N$. Thus $q$ is not equal to any of the $P_i$.
2. Since $N \equiv 3 \pmod 4$, it is not possible that every prime divisor of $N$ is congruent to 1 modulo 4. Indeed, a product of integers each congruent to $1 \pmod 4$ is itself congruent to $1 \pmod 4$ by Lemma Lemma 1.23 (Product of numbers congruent to 1 modulo 4), contradicting $N \equiv 3 \pmod 4$. Hence some prime divisor $q$ of $N$ satisfies $q \equiv 3 \pmod 4$.

Combining the two observations, $q$ is a prime congruent to 3 modulo 4 that is not among $P_1, \ldots, P_r$, contradicting the assumed finiteness of such primes. Therefore there are infinitely many primes congruent to 3 modulo 4. □

**§ Remark (on Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4))** The method above—building an integer with residue $3 \pmod 4$ whose prime divisors cannot all be from a given finite list—gives a proof in this special case of Dirichlet's theorem. However, this argument does not extend directly to all arithmetic progressions; for the full theorem one needs deeper tools (Dirichlet characters and $L$-series).

## Congruences

**§ Historical remark (Gauss and congruences)** Up to the beginning of the nineteenth century number theory consisted of many isolated results. In 1801, Gauss, in his *Disquisitiones Arithmeticae*, introduced the modern theory of congruences and unified many isolated results in number theory.

**§ Definition (Congruence modulo $m$)** Let $a, b, m \in \mathbb{Z}$ with $m > 0$. We say that *$a$ is congruent to $b$ modulo $m$*, and write

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$. The integer $m$ is called the *modulus* of the congruence.

**§ Remark (notation)** The notation $a \not\equiv b \pmod{m}$ means $m \nmid (a - b)$. The modulus is customarily taken to be positive; we will assume $m > 0$ in the sequel.

**§ Proposition 2.1 (Congruence modulo $m$ is an equivalence relation)** Congruence modulo $m$ is reflexive, symmetric, and transitive on $\mathbb{Z}$; hence it is an equivalence relation.

*Proof. Reflexive:* For any $a \in \mathbb{Z}$, $a - a = 0$ and $m \mid 0$, so $a \equiv a \pmod{m}$.

*Symmetric:* If $a \equiv b \pmod{m}$ then $m \mid (a - b)$, hence $m \mid -(a - b) = (b - a)$, so $b \equiv a \pmod{m}$.

*Transitive:* If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $m \mid (a - b)$ and $m \mid (b - c)$, therefore $m \mid (a - b) + (b - c) = a - c$, so $a \equiv c \pmod{m}$. □

**§ Consequence 2.2 (Partition into residue classes)** The set $\mathbb{Z}$ is partitioned into equivalence classes under congruence modulo $m$.

**§ Definition (Notation for residue class)** For $x \in \mathbb{Z}$, write $[x]_m$ (or simply $[x]$ when $m$ is understood) for the equivalence class of $x$ modulo $m$:

$$[x]_m = \{y \in \mathbb{Z} : y \equiv x \pmod{m}\}.$$

(Do not confuse $[x]_m$ with the floor function.)

### Example: The equivalence classes modulo 4

$$\begin{aligned}
[0]_4 &= \{x \in \mathbb{Z} : x \equiv 0 \pmod{4}\} = \{4n : n \in \mathbb{Z}\} = \{\dots, -8, -4, 0, 4, 8, \dots\},\\
[1]_4 &= \{x : x = 4n + 1,\ n \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\},\\
[2]_4 &= \{x : x = 4n + 2,\ n \in \mathbb{Z}\} = \{\dots, -6, -2, 2, 6, 10, \dots\},\\
[3]_4 &= \{x : x = 4n + 3,\ n \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}.
\end{aligned}$$

Thus $\mathbb{Z}$ is partitioned into four residue classes modulo 4, and every integer is congruent to exactly one of $0, 1, 2, 3$ modulo 4.

**§ Definition (Complete residue system modulo $m$)** A set $R \subset \mathbb{Z}$ is a *complete residue system modulo $m$* if every integer is congruent modulo $m$ to exactly one element of $R$.

**§ Example (complete residue systems)** The set $\{0, 1, 2, \dots, m-1\}$ is a complete residue system modulo $m$. Also, for example, $\{6, -11, 19, 1988\}$ is a complete residue system modulo 4, since $6 \equiv 2$, $-11 \equiv 1$, $19 \equiv 3$, and $1988 \equiv 0 \pmod{4}$.

**§ Proposition 2.3 (Standard complete residue system)** The set $\{0, 1, 2, \dots, m-1\}$ is a complete residue system modulo $m$.

*Proof.* Let $a \in \mathbb{Z}$. By the Division Algorithm there exist unique $q, r \in \mathbb{Z}$ with $0 \leq r < m$ such that $a = mq + r$. Then $a - r = mq$, so $m \mid (a - r)$ and $a \equiv r \pmod{m}$, with $r \in \{0, 1, \ldots, m-1\}$. This shows every integer is congruent to at least one element of $\{0, 1, \ldots, m-1\}$.

To show uniqueness: suppose $r_1, r_2 \in \{0, 1, \ldots, m-1\}$ and $r_1 \equiv r_2 \pmod{m}$. Then $m \mid (r_1 - r_2)$, but $-(m-1) \leq r_1 - r_2 \leq m - 1$. The only multiple of $m$ in this range is 0, so $r_1 - r_2 = 0$ and hence $r_1 = r_2$. Thus each integer is congruent to exactly one element of $\{0, 1, \ldots, m-1\}$. $\square$

**§ Proposition 2.4 (Arithmetic with congruences)** Let $a, b, c, d \in \mathbb{Z}$ and $m > 0$. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$,
2. $ac \equiv bd \pmod{m}$.

*Proof.* From $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ we have $m \mid (a - b)$ and $m \mid (c - d)$.

For (1): $m \mid (a - b) + (c - d) = (a + c) - (b + d)$, hence $a + c \equiv b + d \pmod{m}$.

For (2): Note $m \mid (a - b)$ implies $m \mid c(a - b)$, and $m \mid (c - d)$ implies $m \mid b(c - d)$. Add these divisibilities to get $m \mid c(a - b) + b(c - d) = ac - bd$, so $ac \equiv bd \pmod{m}$. $\square$

## 0.6 L6: 2.1 Congruences, 2.2 Linear congruences (1 var.)

# L6: Congruences

**§ Remark (Proposition Proposition 2.4 (Arithmetic with congruences) in class terms)** Recall that Proposition 2.4 (Arithmetic with congruences) from the previous lecture can be expressed in terms of equivalence classes modulo $m$: for residue classes $[a], [c] \in \mathbb{Z}_m$,

$$[a] + [c] = [a + c] \qquad \text{and} \qquad [a][c] = [ac].$$

This viewpoint allows construction of addition and multiplication tables for residue classes (modulo $m$) using any complete residue system.

### Examples: tables for $\mathbb{Z}_4$ and $\mathbb{Z}_5$

| $+$ | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
|---|---|---|---|---|
| $[0]_4$ | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
| $[1]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ | $[0]_4$ |
| $[2]_4$ | $[2]_4$ | $[3]_4$ | $[0]_4$ | $[1]_4$ |
| $[3]_4$ | $[3]_4$ | $[0]_4$ | $[1]_4$ | $[2]_4$ |

| $\cdot$ | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
|---|---|---|---|---|
| $[0]_4$ | $[0]_4$ | $[0]_4$ | $[0]_4$ | $[0]_4$ |
| $[1]_4$ | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
| $[2]_4$ | $[0]_4$ | $[2]_4$ | $[0]_4$ | $[2]_4$ |
| $[3]_4$ | $[0]_4$ | $[3]_4$ | $[2]_4$ | $[1]_4$ |

**§ Remark (structure of $\mathbb{Z}_4$)** Set $\mathbb{Z}_4 := \{[0]_4, [1]_4, [2]_4, [3]_4\}$. With the above addition and multiplication it is a ring, but not a field: not every nonzero element has a multiplicative inverse (e.g., $[2]_4$ is a zero divisor).

| $+$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
|---|---|---|---|---|---|
| $[0]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
| $[1]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ | $[0]_5$ |
| $[2]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ | $[0]_5$ | $[1]_5$ |
| $[3]_5$ | $[3]_5$ | $[4]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ |
| $[4]_5$ | $[4]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ |

| $\cdot$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
|---|---|---|---|---|---|
| $[0]_5$ | $[0]_5$ | $[0]_5$ | $[0]_5$ | $[0]_5$ | $[0]_5$ |
| $[1]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
| $[2]_5$ | $[0]_5$ | $[2]_5$ | $[4]_5$ | $[1]_5$ | $[3]_5$ |
| $[3]_5$ | $[0]_5$ | $[3]_5$ | $[1]_5$ | $[4]_5$ | $[2]_5$ |
| $[4]_5$ | $[0]_5$ | $[4]_5$ | $[3]_5$ | $[2]_5$ | $[1]_5$ |

**§ Remark (structure of $\mathbb{Z}_5$)** Let $\mathbb{Z}_5 := \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$. Because 5 is prime every nonzero class has a multiplicative inverse, so $\mathbb{Z}_5$ is a field.

**§ Question (why extra structure?)** Why does $\mathbb{Z}_5$ have more structure (a field) than $\mathbb{Z}_4$ (only a ring)?

**§ Answer (primality of modulus)** Essentially: $\mathbb{Z}_m$ is a field if and only if $m$ is prime. If $m$ is composite, then some nonzero classes are zero divisors (no inverses). If $m = p$ is prime, then every nonzero class has an inverse (standard proof via Bezout's identity).

### Cancellation and a useful proposition

**§ Proposition 2.5 (Cancellation in congruences)** Let $a, b, c, m \in \mathbb{Z}$ with $m > 0$, and let $d = \gcd(c, m)$. Then

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{m/d}.$$

*Proof.* Suppose $ca \equiv cb \pmod{m}$, i.e. $m \mid c(a-b)$. Let $d = \gcd(c, m)$. Write $c = d \cdot c'$ and $m = d \cdot m'$ so that $\gcd(c', m') = 1$. From $m \mid c(a-b)$ we get $dm' \mid dc'(a-b)$, hence $m' \mid c'(a-b)$. Since $\gcd(c', m') = 1$, it follows that $m' \mid (a-b)$, i.e. $a \equiv b \pmod{m'}$, which is $a \equiv b \pmod{m/d}$.

Conversely, if $a \equiv b \pmod{m/d}$ then $m/d \mid (a-b)$, so $m/d \cdot d = m \mid d(a-b)$. But $d \mid c$, so $c(a-b)$ is a multiple of $d(a-b)$, hence $m \mid c(a-b)$ and $ca \equiv cb \pmod{m}$. This proves the equivalence. □

## Linear congruences in one variable

**§ Definition (linear congruence)** A congruence of the form

$$ax \equiv b \pmod{m},$$

with unknown $x \in \mathbb{Z}$ and given integers $a, b, m$ with $m > 0$, is called a *linear congruence* in the variable $x$.

**§ Examples (small checks)**

- $2x \equiv 3 \pmod 4$ has no solution: checking $x \equiv 0, 1, 2, 3 \pmod 4$ yields no solution.
- $2x \equiv 3 \pmod 5$ has the unique solution $x \equiv 4 \pmod 5$.
- $2x \equiv 4 \pmod 6$ has two incongruent solutions modulo 6: $x \equiv 2, 5 \pmod 6$.
- $3x \equiv 9 \pmod 6$ has three incongruent solutions modulo 6: $x \equiv 1, 3, 5 \pmod 6$.

**§ Theorem 2.6 (Solvability and number of solutions of linear congruences)** Let $a, b, m \in \mathbb{Z}$ with $m > 0$ and let $d = \gcd(a, m)$. Then:

1. The congruence $ax \equiv b \pmod m$ has a solution iff $d \mid b$.
2. If $d \mid b$ then there are exactly $d$ incongruent solutions modulo $m$. Concretely, if $x_0$ is one solution, then all solutions are
$$x = x_0 + \frac{m}{d}n, \qquad n \in \mathbb{Z},$$
and the classes $x_0 + \frac{m}{d}n$ for $n = 0, 1, \ldots, d-1$ give a complete set of incongruent solutions modulo $m$.

*Proof.* We use the standard equivalence between the congruence and a linear Diophantine equation. The congruence $ax \equiv b \pmod m$ is equivalent to the existence of an integer $y$ such that

$$ax - my = b. \tag{*}$$

Thus solvability of the congruence is equivalent to solvability of the linear Diophantine equation $(*)$.

(1) If $(*)$ has an integer solution, then $d = \gcd(a, m)$ divides the left-hand side $ax - my$ and hence divides $b$. Conversely, if $d \mid b$, write $b = de$ for some $e \in \mathbb{Z}$. By Bezout's identity (see **??** or the standard result), there exist integers $r, s$ with $ar + ms = d$. Multiply both sides by $e$ to obtain $are + mse = de = b$. Then $x_0 = re$ and $y_0 = -se$ solve $(*)$, so a solution exists.

(2) Fix any solution $x_0$ of $ax \equiv b \pmod m$. For any integer $n$ consider

$$x = x_0 + \frac{m}{d}n.$$

Then

$$ax = ax_0 + a\frac{m}{d}n = ax_0 + \frac{a}{d}mn \equiv ax_0 \equiv b \pmod{m},$$

so each such $x$ is a solution. Thus there are infinitely many integer solutions, all congruent modulo $\frac{m}{d}$ shifts.

Now suppose $x$ is any solution. Then subtracting the equation for $x_0$ from the equation for $x$ yields

$$a(x - x_0) \equiv 0 \pmod{m},$$

i.e. $m \mid a(x - x_0)$. Write $a = da'$ and $m = dm'$, with $\gcd(a', m') = 1$. Then $m'd \mid da'(x - x_0)$, so $m' \mid a'(x - x_0)$. Since $\gcd(a', m') = 1$ we deduce $m' \mid (x - x_0)$, hence $x - x_0 = m'n = \frac{m}{d}n$ for some $n \in \mathbb{Z}$. This shows every solution has the stated form. Two such solutions

$$x_0 + \frac{m}{d}n_1 \quad \text{and} \quad x_0 + \frac{m}{d}n_2$$

are congruent modulo $m$ iff

$$\frac{m}{d}(n_1 - n_2) \equiv 0 \pmod{m} \iff m \mid \frac{m}{d}(n_1 - n_2).$$

Dividing both sides by $m/d$ yields $d \mid (n_1 - n_2)$, so $n_1 \equiv n_2 \pmod{d}$. Therefore choosing $n = 0, 1, \ldots, d-1$ produces exactly $d$ incongruent solutions modulo $m$. □

**§ Corollary 2.7 (Explicit solution set)** Under the hypotheses of Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences), if $d = \gcd(a, m)$ and $d \mid b$, then the $d$ incongruent solutions modulo $m$ are

$$x_0 + \frac{m}{d}n, \qquad n = 0, 1, \ldots, d-1,$$

where $x_0$ is any particular solution.

## Worked example

**§ Example (solve $16x \equiv 8 \pmod{28}$)** We solve $16x \equiv 8 \pmod{28}$. Compute $d = \gcd(16, 28)$.

Euclidean algorithm:

$$28 = 1 \cdot 16 + 12, \qquad 16 = 1 \cdot 12 + 4, \qquad 12 = 3 \cdot 4 + 0,$$

so $d = 4$. Since $4 \mid 8$, the congruence has $d = 4$ incongruent solutions modulo 28 by Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences).

One method: divide the congruence by $d = 4$ to get (working modulo $28/4 = 7$)

$$4x \equiv 2 \pmod{7}.$$

Since $4^{-1} \equiv 2 \pmod{7}$ (because $4 \cdot 2 = 8 \equiv 1 \pmod{7}$), multiply both sides by 2:

$$x \equiv 4 \pmod{7}.$$

Thus the solutions modulo 28 are

$$x \equiv 4 + 7n, \qquad n = 0, 1, 2, 3,$$

which gives the incongruent solutions $x \equiv 4, 11, 18, 25 \pmod{28}$.

(Alternatively, one can find an explicit $x_0$ via the extended Euclidean computation: from $4 = 2 \cdot 16 - 1 \cdot 28$ we get $8 = 4 \cdot 16 - 2 \cdot 28$, so $x_0 = 4$ is a particular solution, yielding the same solution set.)

## Multiplicative inverses

**§ Definition (multiplicative inverse modulo $m$)** If $a \in \mathbb{Z}$ and there exists $x \in \mathbb{Z}$ with $ax \equiv 1 \pmod{m}$, then $x$ is called a *multiplicative inverse* of $a$ modulo $m$ (often denoted $a^{-1} \pmod{m}$).

**§ Corollary 2.8 (When inverses exist)** The congruence $ax \equiv 1 \pmod{m}$ has a solution iff $\gcd(a, m) = 1$. In that case the inverse is unique modulo $m$.

*Proof.* Apply Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences) with $b = 1$. If $d = \gcd(a, m) > 1$ then $d \nmid 1$ so there is no solution. If $d = 1$ then there is exactly one solution modulo $m$. □

**§ Example (solve $5x \equiv 12 \pmod{16}$)** Since $\gcd(5, 16) = 1$, the congruence has a unique solution modulo 16 by Corollary Corollary 2.8 (When inverses exist). One checks that $5^{-1} \pmod{16} = 13$ because $5 \cdot 13 = 65 \equiv 1 \pmod{16}$. Multiplying by 13 gives

$$x \equiv 13 \cdot 12 \pmod{16}.$$

Compute $13 \cdot 12 = 156 \equiv 12 \pmod{16}$ (since $156 - 144 = 12$), so $x \equiv 12 \pmod{16}$.

## 0.7 L7: 2.3 Chinese Remainder Thm, 2.4 Wilson's Thm, 2.5 Fermat's Little Thm

# L7: The Chinese Remainder Theorem, Wilson's Theorem, Fermat's Little Theorem

**Example (motivation).** Find a positive integer $x$ with

$$x \equiv 2 \pmod 3, \qquad x \equiv 1 \pmod 4, \qquad x \equiv 3 \pmod 5.$$

This is a system of linear congruences in one variable.

> **§ Theorem 2.9 (Chinese Remainder Theorem)** Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers, and let $b_1, \dots, b_n$ be arbitrary integers. Then the system
>
> $$x \equiv b_i \pmod{m_i} \qquad (i = 1, \dots, n)$$
>
> has a solution, and any two solutions are congruent modulo $M := m_1 m_2 \cdots m_n$. Hence the solution is unique modulo $M$.

*Proof.* Let $M := \prod_{i=1}^n m_i$ and define $M_i := M/m_i$. Since $(M_i, m_i) = 1$ by the pairwise coprimality hypothesis, by Corollary 2.8 (When inverses exist) (existence of inverses when gcd is 1) there exists $x_i$ with

$$M_i x_i \equiv 1 \pmod{m_i}.$$

Put

$$x := \sum_{i=1}^n b_i M_i x_i.$$

For a fixed index $i$, reduce modulo $m_i$: every term $b_j M_j x_j$ with $j \neq i$ is a multiple of $m_i$ (since $m_i \mid M_j$ for $j \neq i$), while $M_i x_i \equiv 1 \pmod{m_i}$ by construction. Thus $x \equiv b_i \pmod{m_i}$ for each $i$, so $x$ is a solution.

If $x'$ is another solution, then $m_i \mid (x - x')$ for every $i$, and since the $m_i$ are pairwise coprime it follows that $M \mid (x - x')$. Hence $x \equiv x' \pmod M$, proving uniqueness modulo $M$. □

**Example (continued).** Here $m_1 = 3, m_2 = 4, m_3 = 5$, so $M = 60$ and

$$M_1 = 60/3 = 20, \qquad M_2 = 60/4 = 15, \qquad M_3 = 60/5 = 12.$$

Solve $M_i x_i \equiv 1 \pmod{m_i}$:

$$20x_1 \equiv 1 \pmod 3 \implies 2x_1 \equiv 1 \pmod 3 \implies x_1 \equiv 2 \pmod 3,$$

so choose $x_1 = 2$.

$$15x_2 \equiv 1 \pmod 4 \implies 3x_2 \equiv 1 \pmod 4 \implies x_2 \equiv 3 \pmod 4,$$

so choose $x_2 = 3$.

$$12x_3 \equiv 1 \pmod 5 \implies 2x_3 \equiv 1 \pmod 5 \implies x_3 \equiv 3 \pmod 5,$$

so choose $x_3 = 3$.

Now

$$x = b_1 M_1 x_1 + b_2 M_2 x_2 + b_3 M_3 x_3 = 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 3 \cdot 12 \cdot 3 = 80 + 45 + 108 = 233.$$

Hence $x \equiv 233 \equiv 53 \pmod{60}$, and the least positive solution is 53.

**§ Lemma 2.10 (Self-inverse modulo a prime)** Let $p$ be a prime and $a \in \mathbb{Z}$. Then $a$ is its own inverse modulo $p$ (i.e. $a^2 \equiv 1 \pmod{p}$) if and only if $a \equiv \pm 1 \pmod{p}$.

*Proof.* If $a^2 \equiv 1 \pmod{p}$ then $p \mid (a-1)(a+1)$. Since $p$ is prime, $p \mid (a-1)$ or $p \mid (a+1)$, so $a \equiv 1$ or $a \equiv -1 \pmod{p}$. The converse is immediate. □

**§ Theorem 2.11 (Wilson's Theorem)** Let $p$ be a prime. Then

$$(p-1)! \equiv -1 \pmod{p}.$$

*Proof.* The result is trivial for $p = 2, 3$, so assume $p \geq 5$. The integers $1, 2, \ldots, p-1$ can be paired into multiplicative inverses modulo $p$. By Lemma 0.7, the only elements that are their own inverses are 1 and $p-1$. Thus, aside from 1 and $p-1$, the remaining $p-3$ elements pair off, and each pair multiplies to 1 modulo $p$. Therefore

$$(p-1)! \equiv 1 \cdot (\text{product of inverse-pairs}) \cdot (p-1) \equiv 1 \cdot 1 \cdots 1 \cdot (-1) \equiv -1 \pmod{p},$$

as required. □

**§ Remark (converse)** The converse is also true: if $n > 1$ and $(n-1)! \equiv -1 \pmod{n}$ then $n$ must be prime.

**§ Proposition 2.12 (Converse to Wilson's Theorem)** Let $n > 1$ be an integer. If $(n-1)! \equiv -1 \pmod{n}$ then $n$ is prime.

*Proof.* Suppose $n$ is composite, so $n = ab$ with $1 < a \leq b < n$. Then $a \mid (n-1)!$ because $(n-1)!$ contains the factor $a$, and hence $(n-1)! \equiv 0 \pmod{a}$. The hypothesis $(n-1)! \equiv -1 \pmod{n}$ implies $(n-1)! \equiv -1 \pmod{a}$ (since $a \mid n$), so $-1 \equiv 0 \pmod{a}$, which is impossible. Therefore no such nontrivial factor $a$ exists and $n$ must be prime. □

**§ Remark (Wilson primes)** A prime $p$ is called a *Wilson prime* if $(p-1)! \equiv -1 \pmod{p^2}$. Only three Wilson primes are known: $5, 13, 563$. It is unknown whether infinitely many (or only finitely many) Wilson primes exist.

**§ Theorem 2.13 (Fermat's Little Theorem)** Let $p$ be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

*Proof.* Consider the $p-1$ integers $a, 2a, \ldots, (p-1)a$. None of these is congruent to $0 \pmod{p}$ (since $p \nmid a$), and no two are congruent modulo $p$ (if $ia \equiv ja \pmod{p}$ with $1 \leq i < j \leq p-1$, multiply by the inverse of $a$ modulo $p$, which exists by Corollary 2.8 (When inverses exist), to deduce $i \equiv j \pmod{p}$, impossible). Hence the residues of $a, 2a, \ldots, (p-1)a$ modulo $p$ are a permutation of $1, 2, \ldots, p-1$. Multiplying all these congruences yields

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since $p \nmid (p-1)!$ (true for prime $p$), we may cancel $(p-1)!$ from both sides to obtain $a^{p-1} \equiv 1 \pmod{p}$. (Note: this cancellation step uses only that $\gcd((p-1)!, p) = 1$; Wilson's Theorem is not required.) □

### 0.8 L8: 2.5 Pseudoprimes, 2.6 Euler's Thm, 3.1 Arithmetic functions/multiplicativity

## L8: Corollaries to Fermat's Little Theorem, Pseudoprimes, Euler's Theorem, Arithmetic Functions

**§ Corollary 2.14 (Inverse modulo a prime)** Let $p$ be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then $a^{p-2}$ is the multiplicative inverse of $a$ modulo $p$; that is,

$$a \cdot a^{p-2} \equiv 1 \pmod{p}.$$

*Proof.* By Fermat's Little Theorem (Theorem 0.7), $a^{p-1} \equiv 1 \pmod{p}$. Multiply both sides by $a^{-1}$ if you like, or simply rewrite $a^{p-1} = a \cdot a^{p-2}$ to get $a \cdot a^{p-2} \equiv 1 \pmod{p}$. Thus $a^{p-2}$ is an inverse of $a$ modulo $p$. □

**§ Remark (application of Corollary Corollary 2.14 (Inverse modulo a prime))** Corollary Corollary 2.14 (Inverse modulo a prime) is useful for solving linear congruences $ax \equiv b \pmod{p}$ when $p$ is prime and $p \nmid a$: multiply both sides by $a^{p-2}$ to obtain the solution.

**§ Corollary 2.15 (Fresh form of Fermat)** Let $p$ be prime and $a \in \mathbb{Z}$. Then

$$a^p \equiv a \pmod{p}.$$

*Proof.* If $p \mid a$ the congruence is trivial. If $p \nmid a$, multiply Fermat's Little Theorem $a^{p-1} \equiv 1 \pmod{p}$ by $a$ to get $a^p \equiv a \pmod{p}$. □

**§ Corollary 2.16 (case $a = 2$)** For every prime $p$,

$$2^p \equiv 2 \pmod{p}.$$

*Proof.* Take $a = 2$ in Corollary Corollary 2.15 (Fresh form of Fermat). □

**§ Question (converse for base 2)** Is the converse of Corollary Corollary 2.16 (case $a = 2$) true? That is, if $n > 1$ and $2^n \equiv 2 \pmod{n}$, must $n$ be prime?

**§ Answer and counterexample (341)** No. Example: $n = 341 = 11 \cdot 31$ is composite but satisfies $2^{341} \equiv 2 \pmod{341}$. Since $(11, 31) = 1$, it suffices to check modulo 11 and 31.

By Fermat's Little Theorem,

$$2^{10} \equiv 1 \pmod{11} \quad \Rightarrow \quad 2^{341} \equiv (2^{10})^{34} \cdot 2 \equiv 1^{34} \cdot 2 \equiv 2 \pmod{11},$$

and

$$2^{30} \equiv 1 \pmod{31} \quad \Rightarrow \quad 2^{341} \equiv (2^{30})^{11} \cdot 2^{11} \equiv 1^{11} \cdot 2^{11} \pmod{31}.$$

Since $2^5 \equiv -1 \pmod{31}$ (or compute directly), we have $2^{11} \equiv 2 \pmod{31}$, so $2^{341} \equiv 2 \pmod{31}$. Hence $2^{341} \equiv 2 \pmod{341}$ although 341 is composite.

**§ Definition (pseudoprime to base 2)** A composite integer $n > 1$ is called a *(base-2) pseudoprime* if

$$2^n \equiv 2 \pmod{n}.$$

**§ Remark (pseudoprimes)** 341 is the smallest base-2 pseudoprime. There are infinitely many pseudoprimes and they arise for different bases; pseudoprimes show that the naive converse of Fermat's Little Theorem is false.

**§ Definition (Euler's phi-function)** For $n \in \mathbb{Z}_{>0}$ the Euler totient function $\phi(n)$ is the number of integers $1 \le x \le n$ with $\gcd(x, n) = 1$; equivalently,

$$\phi(n) := \#\{x \in \{1, 2, \ldots, n\} : \gcd(x, n) = 1\}.$$

**§ Examples (values of $\phi$)**

$$\phi(12) = 4 \quad \text{(the classes } 1, 5, 7, 11\text{)}, \qquad \phi(14) = 6 \quad \text{(the classes } 1, 3, 5, 9, 11, 13\text{)},$$

and for a prime $p$, $\phi(p) = p - 1$ (classes $1, 2, \ldots, p - 1$).

**§ Theorem 2.17 (Euler's Theorem)** Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

*Proof.* Let $r_1, r_2, \ldots, r_{\phi(m)}$ be the positive integers $\le m$ that are relatively prime to $m$. Since $\gcd(a, m) = 1$, multiplication by $a$ permutes this reduced residue system: the numbers $ar_1, \ldots, ar_{\phi(m)}$ are all distinct modulo $m$ and each is relatively prime to $m$ (see Lemma 1.5 and Lemma 1.14 referenced earlier). Hence the set $\{ar_i \bmod m\}$ is the same set of residues as $\{r_i\}$. Multiplying all residues yields

$$a^{\phi(m)} \prod_{i=1}^{\phi(m)} r_i \equiv \prod_{i=1}^{\phi(m)} r_i \pmod{m}.$$

Since $\gcd\left(\prod_i r_i,\, m\right) = 1$, we may cancel the product to obtain $a^{\phi(m)} \equiv 1 \pmod{m}$. □

**§ Remark (special case)** Fermat's Little Theorem is the special case $m = p$ prime, since $\phi(p) = p - 1$.

**§ Definition (reduced residue system)** A set of $\phi(m)$ integers each relatively prime to $m$ and pairwise incongruent modulo $m$ is called a *reduced residue system* modulo $m$.

**§ Example (reduced residue systems)** $\{1, 5, 7, 11\}$ is a reduced residue system modulo 12. If $a$ is coprime to $m$, then $\{ar_1, \ldots, ar_{\phi(m)}\}$ is also a reduced residue system modulo $m$ (this was the idea used in the proof of Euler's Theorem).

**§ Corollary 2.19 (inverse via Euler)** Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then $a^{\phi(m)-1}$ is the multiplicative inverse of $a$ modulo $m$.

*Proof.* From $a^{\phi(m)} \equiv 1 \pmod{m}$ multiply by $a^{-1}$ (or rewrite as $a \cdot a^{\phi(m)-1} \equiv 1$) to see $a^{\phi(m)-1}$ is an inverse of $a$ modulo $m$. □

**§ Remark (use of Corollary Corollary 2.19 (inverse via Euler))** Corollary Corollary 2.19 (inverse via Euler) helps solve linear congruences $ax \equiv b \pmod{m}$ when $\gcd(a, m) = 1$: multiply both sides by $a^{\phi(m)-1}$ to obtain the solution.

**§ Definition (arithmetic function)** An *arithmetic function* is a function $f : \mathbb{Z}_{>0} \to \mathbb{C}$ (or $\mathbb{R}$, or $\mathbb{Z}$). Examples: the Euler $\phi$-function, the divisor-counting function $d(n)$, the sum-of-divisors function $\sigma(n)$, etc.

**§ Definition (multiplicative vs completely multiplicative)** An arithmetic function $f$ is *multiplicative* if $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is *completely multiplicative* if $f(mn) = f(m)f(n)$ for all positive integers $m, n$ (no coprimality required).

**§ Proposition (values determined by prime powers)** If $f$ is multiplicative and

$$n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$$

is the prime factorization of $n$, then

$$f(n) = \prod_{i=1}^{r} f\big(p_i^{a_i}\big).$$

If $f$ is completely multiplicative, then

$$f(n) = \prod_{i=1}^{r} f(p_i)^{a_i}.$$

*Proof.* Write $n$ as a product of relatively prime prime-power factors and apply multiplicativity repeatedly. $\square$

**§ Remark (implication)** Thus knowing a multiplicative arithmetic function on prime powers (or a completely multiplicative function on primes) determines it everywhere.

## 0.9 L9: 3.1 Arithmetic functions/multiplicativity, 3.2 Euler phi function

# L9: Arithmetic Functions and Multiplicativity

**§ Summation over divisors: notation** By

$$\sum_{d|n} f(d)$$

we mean the sum of $f(d)$ over all *positive* divisors $d$ of $n$. For example,

$$\sum_{d|12} f(d) = f(1) + f(2) + f(3) + f(4) + f(6) + f(12).$$

**§ Multiplicative functions and divisor sums** Let $f$ be an arithmetic function and define, for $n \in \mathbb{Z}_{>0}$,

$$F(n) := \sum_{d|n} f(d).$$

If $f$ is multiplicative then $F$ is multiplicative.

*Proof.* Let $m, n \in \mathbb{Z}_{>0}$ with $\gcd(m, n) = 1$. We must show $F(mn) = F(m)\,F(n)$.

Claim: Every positive divisor $d$ of $mn$ can be written uniquely as $d = d_1 d_2$ with $d_1 \mid m$, $d_2 \mid n$ and $\gcd(d_1, d_2) = 1$.

Proof of claim: Write the prime factorizations

$$m = \prod_{i=1}^{r} p_i^{a_i}, \qquad n = \prod_{j=1}^{s} q_j^{b_j},$$

where the primes $p_i$ and $q_j$ are distinct because $\gcd(m, n) = 1$. Then any divisor $d$ of $mn$ has the form

$$d = \prod_{i=1}^{r} p_i^{e_i} \prod_{j=1}^{s} q_j^{f_j},$$

with $0 \le e_i \le a_i$ and $0 \le f_j \le b_j$. Set

$$d_1 = \prod_{i=1}^{r} p_i^{e_i}, \qquad d_2 = \prod_{j=1}^{s} q_j^{f_j}.$$

Clearly $d_1 \mid m$, $d_2 \mid n$, $\gcd(d_1, d_2) = 1$, and $d = d_1 d_2$. Uniqueness follows from uniqueness of prime factorization. □

Using the claim,

$$F(mn) = \sum_{d|mn} f(d) = \sum_{\substack{d_1|m \\ d_2|n}} f(d_1 d_2).$$

Since $f$ is multiplicative and $\gcd(d_1, d_2) = 1$ for every pair occurring in the sum, we have $f(d_1 d_2) = f(d_1)f(d_2)$. Thus

$$F(mn) = \sum_{d_1|m} \sum_{d_2|n} f(d_1)f(d_2) = \Big(\sum_{d_1|m} f(d_1)\Big)\Big(\sum_{d_2|n} f(d_2)\Big) = F(m)F(n).$$

This proves that $F$ is multiplicative. □

**§ Example: verification for $m = 3$, $n = 4$** Let $m = 3$ and $n = 4$. The positive divisors are

$$\{\, d_1 \mid 3 \,\} = \{1, 3\}, \qquad \{\, d_2 \mid 4 \,\} = \{1, 2, 4\}.$$

All divisors of 12 are the products $d_1 d_2$ with $d_1 \in \{1,3\}$ and $d_2 \in \{1,2,4\}$; hence

$$\sum_{d|12} f(d) = \sum_{d_1|3}\sum_{d_2|4} f(d_1 d_2) = \sum_{d_1|3}\sum_{d_2|4} f(d_1)f(d_2)$$

(using multiplicativity of $f$), so

$$\sum_{d|12} f(d) = \Big(\sum_{d_1|3} f(d_1)\Big)\Big(\sum_{d_2|4} f(d_2)\Big) = F(3)F(4).$$

**§ The Euler $\varphi$-function is multiplicative** The Euler totient function $\varphi(n)$ (the number of integers in $\{1,2,\ldots,n\}$ relatively prime to $n$) is multiplicative: if $\gcd(m,n)=1$ then

$$\varphi(mn) = \varphi(m)\,\varphi(n).$$

*Proof.* List the integers $1,2,\ldots,mn$ in an $m \times n$ array with rows indexed by $i = 1,\ldots,m$ and columns by $k = 0,\ldots,n-1$ as

$$\begin{array}{ccccc} 1 & m+1 & 2m+1 & \cdots & (n-1)m+1 \\ 2 & m+2 & 2m+2 & \cdots & (n-1)m+2 \\ \vdots & \vdots & \vdots & & \vdots \\ m & m+m & 2m+m & \cdots & (n-1)m+m \end{array}$$

Consider the $i$-th row

$$i,\ m+i,\ 2m+i,\ \ldots,\ (n-1)m+i.$$

If $\gcd(i,m) > 1$ then every entry of that row shares a common factor with $m$, and hence cannot be relatively prime to $mn$. Thus only rows with $\gcd(i,m)=1$ can contain integers relatively prime to $mn$. There are precisely $\varphi(m)$ such rows.

Fix a row with $\gcd(i,m)=1$. The $n$ entries of that row are distinct modulo $n$: if $jm+i \equiv km+i \pmod n$ with $0 \le j,k \le n-1$, then $(j-k)m \equiv 0 \pmod n$. Since $\gcd(m,n)=1$, $m$ has a multiplicative inverse modulo $n$, so $j \equiv k \pmod n$, hence $j=k$. Thus the row is a complete residue system modulo $n$, so exactly $\varphi(n)$ of its entries are relatively prime to $n$. Because those entries are relatively prime to both $m$ and $n$, they are relatively prime to $mn$.

Therefore each of the $\varphi(m)$ admissible rows contains exactly $\varphi(n)$ integers relatively prime to $mn$. Thus

$$\varphi(mn) = \varphi(m)\varphi(n).$$

□

**§ Values of $\varphi$ at prime powers** Let $p$ be prime and $a \in \mathbb{Z}_{>0}$. Then

$$\varphi(p^a) = p^a - p^{a-1} = p^a\Big(1 - \frac{1}{p}\Big).$$

*Proof.* Among the $p^a$ integers $1,2,\ldots,p^a$, those not coprime to $p^a$ are exactly the multiples of $p$:

$$p, 2p, 3p, \ldots, p^{a-1}p,$$

and there are $p^{a-1}$ such multiples. Hence

$$\varphi(p^a) = p^a - p^{a-1} = p^a\Big(1 - \frac{1}{p}\Big).$$

□

**§ Product formula for $\varphi(n)$** For $n \in \mathbb{Z}_{>0}$,

$$\varphi(n) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big),$$

where the product is taken over the distinct prime divisors $p$ of $n$.

*Proof.* If $n = 1$ the result holds (empty product $= 1$). For $n > 1$, write the prime factorization

$$n = \prod_{i=1}^{r} p_i^{a_i}.$$

By Theorems The Euler $\varphi$-function is multiplicative and Values of $\varphi$ at prime powers,

$$\varphi(n) = \prod_{i=1}^{r} \varphi(p_i^{a_i}) = \prod_{i=1}^{r} p_i^{a_i}\Big(1 - \frac{1}{p_i}\Big) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big).$$

□

**§ Example: $\varphi(504)$** Since $504 = 2^3 \cdot 3^2 \cdot 7$, we get

$$\varphi(504) = 504\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{7}\Big).$$

Compute:

$$\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{7}\Big) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{6}{7} = \frac{2}{7},$$

so $\varphi(504) = 504 \cdot \frac{2}{7} = 144$.

**§ Gauss's identity for $\varphi$** For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d \mid n} \varphi(d) = n.$$

*Proof.* For each positive divisor $d$ of $n$ define

$$S_d := \{\, m \in \mathbb{Z} : 1 \le m \le n,\ \gcd(m, n) = d \,\}.$$

Note that $\gcd(m, n) = d$ if and only if $\gcd(m/d, n/d) = 1$, so the number of integers in $S_d$ equals $\varphi(n/d)$. The sets $S_d$ (as $d$ runs over the positive divisors of $n$) partition the set $\{1, 2, \dots, n\}$; hence

$$n = \sum_{d \mid n} |S_d| = \sum_{d \mid n} \varphi(n/d).$$

Reindexing the sum by replacing $d$ with $n/d$ shows

$$n = \sum_{d \mid n} \varphi(d),$$

as claimed. □

**§ Example: $n = 12$** For $n = 12$ the divisors and corresponding sets are

$$\begin{aligned}
S_1 &= \{1, 5, 7, 11\}, & |S_1| &= \varphi(12) = 4, \\
S_2 &= \{2, 10\}, & |S_2| &= \varphi(6) = 2, \\
S_3 &= \{3, 9\}, & |S_3| &= \varphi(4) = 2, \\
S_4 &= \{4, 8\}, & |S_4| &= \varphi(3) = 2, \\
S_6 &= \{6\}, & |S_6| &= \varphi(2) = 1, \\
S_{12} &= \{12\}, & |S_{12}| &= \varphi(1) = 1.
\end{aligned}$$

Summing gives

$$\sum_{d|12} \varphi(d) = 4 + 2 + 2 + 2 + 1 + 1 = 12,$$

confirming Gauss's identity.

## 0.10 L10: 3.3 Divisor function, 3.4 Sum of divisors, 3.5 Perfect numbers, 3.6 Möbius function

# L10: The number of positive divisors function

**§ The number of positive divisors function** Let $n \in \mathbb{Z}_{>0}$. The number of positive divisors of $n$ (the divisor-count function) is the function

$$V(n) := \#\{\, d \in \mathbb{Z}_{>0} : d \mid n \,\}.$$

**§ Multiplicativity of the divisor-count function** The function $V(n)$ is multiplicative: if $\gcd(m, n) = 1$ then

$$V(mn) = V(m)\, V(n).$$

*Proof.* Write

$$V(n) = \sum_{d \mid n} 1,$$

the sum of the constant arithmetic function $f(d) \equiv 1$ over positive divisors of $n$. The constant function $f(d) = 1$ is multiplicative, so by Theorem Multiplicative functions and divisor sums (divisor-sum preserving multiplicativity) the divisor-sum $V(n)$ is multiplicative. □

**§ Values of $V$ at prime powers** Let $p$ be prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$V\left(p^a\right) = a + 1.$$

*Proof.* The positive divisors of $p^a$ are exactly

$$1, p, p^2, \ldots, p^a,$$

so there are $a + 1$ of them. □

**§ Prime-power formula for $V(n)$** If

$$n = \prod_{i=1}^{r} p_i^{a_i}$$

is the prime factorization of $n$ (distinct primes $p_i$, exponents $a_i \geq 0$), then

$$V(n) = \prod_{i=1}^{r} (a_i + 1).$$

*Proof.* Since $V$ is multiplicative (Theorem Multiplicativity of the divisor-count function), and each $V(p_i^{a_i}) = a_i + 1$ (Theorem Values of $V$ at prime powers), multiplicativity yields

$$V(n) = \prod_{i=1}^{r} V(p_i^{a_i}) = \prod_{i=1}^{r} (a_i + 1).$$

□

**§ Example: $V(504)$** If $504 = 2^3 \cdot 3^2 \cdot 7^1$ then

$$V(504) = (3 + 1)(2 + 1)(1 + 1) = 4 \cdot 3 \cdot 2 = 24.$$

**§ The sum-of-divisors function $\sigma$** For $n \in \mathbb{Z}_{>0}$ define

$$\sigma(n) := \sum_{d \mid n} d,$$

the sum of the positive divisors of $n$.

**§ Multiplicativity of $\sigma$** The function $\sigma(n)$ is multiplicative: if $\gcd(m,n) = 1$ then

$$\sigma(mn) = \sigma(m)\,\sigma(n).$$

*Proof.* This follows from Theorem Multiplicative functions and divisor sums by observing that the arithmetic function $f(d) = d$ is multiplicative: if $\gcd(a,b) = 1$ then $(ab) = a \cdot b$ and summing multiplicatively gives the result. (Equivalently, check $f(ab) = ab = f(a)f(b)$ for coprime $a, b$.) □

**§ Values of $\sigma$ at prime powers** Let $p$ be prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$\sigma\big(p^a\big) = 1 + p + \cdots + p^a = \frac{p^{a+1} - 1}{p - 1}.$$

*Proof.* The divisors of $p^a$ are $1, p, \ldots, p^a$, so the sum is the finite geometric series $1 + p + \cdots + p^a = (p^{a+1} - 1)/(p-1)$. □

**§ Prime-power product formula for $\sigma(n)$** If $n = \prod_{i=1}^{r} p_i^{a_i}$ then

$$\sigma(n) = \prod_{i=1}^{r} \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

*Proof.* By multiplicativity (Theorem Multiplicativity of $\sigma$) and Theorem Values of $\sigma$ at prime powers applied to each prime power factor. □

**§ Example: $\sigma(504)$** For $504 = 2^3 \cdot 3^2 \cdot 7$,

$$\sigma(504) = \Big(\frac{2^4 - 1}{2 - 1}\Big)\Big(\frac{3^3 - 1}{3 - 1}\Big)\Big(\frac{7^2 - 1}{7 - 1}\Big) = 15 \cdot 13 \cdot 8 = 1560.$$

**§ Perfect numbers (definition)** A positive integer $n$ is *perfect* if $\sigma(n) = 2n$. Equivalently, the sum of the proper divisors of $n$ (i.e., divisors $< n$) equals $n$.

**§ Classical characterization of even perfect numbers** An integer $n > 0$ is an even perfect number if and only if

$$n = 2^{p-1}\big(2^p - 1\big),$$

where $p$ is prime and $2^p - 1$ is prime (a Mersenne prime).

*Proof.* We give both directions (Euler $\Rightarrow$ and Euclid $\Leftarrow$).

($\Leftarrow$) **(Euclid).** Suppose $p$ and $2^p - 1$ are prime and set $n = 2^{p-1}(2^p - 1)$. Using multiplicativity of $\sigma$,

$$\sigma(n) = \sigma\big(2^{p-1}\big)\sigma(2^p - 1) = \frac{2^p - 1}{2 - 1} \cdot (1 + 2^p - 1) = (2^p - 1) \cdot 2^p = 2n,$$

since $\sigma(2^{p-1}) = 1 + 2 + \cdots + 2^{p-1} = (2^p - 1)$ and $\sigma(2^p - 1) = 1 + (2^p - 1) = 2^p$ (because $2^p - 1$ is prime). Hence $n$ is perfect.

($\Rightarrow$) **(Euler).** Let $n$ be an even perfect number. Write $n = 2^r m$ with $r \geq 1$ and $m$ odd. Then multiplicativity yields

$$\sigma(n) = \sigma(2^r)\sigma(m) = (2^{r+1} - 1)\sigma(m).$$

Since $n$ is perfect, $\sigma(n) = 2n = 2^{r+1}m$. Thus

$$(2^{r+1} - 1)\sigma(m) = 2^{r+1}m.$$

Because $\gcd(2^{r+1} - 1, 2^{r+1}) = 1$, it follows that $2^{r+1} \mid \sigma(m)$. Hence write $\sigma(m) = 2^{r+1}c$ for some $c \in \mathbb{Z}_{>0}$, and substituting back gives

$$(2^{r+1} - 1)2^{r+1}c = 2^{r+1}m \quad \Longrightarrow \quad m = (2^{r+1} - 1)c.$$

We now show $c = 1$. If $c > 1$ then $m$ has at least the divisors $1, c, m$ with $1 < c < m$, so

$$\sigma(m) \geq 1 + c + m = 1 + c + (2^{r+1} - 1)c = 1 + (2^{r+1})c,$$

whence $\sigma(m) \geq 1 + 2^{r+1}c$, contradicting $\sigma(m) = 2^{r+1}c$. Thus $c = 1$ and $m = 2^{r+1} - 1$. Then $\sigma(m) = m + 1 = 2^{r+1}$, so $m$ is prime and $m = 2^{r+1} - 1$ is a Mersenne prime. Setting $p = r + 1$ gives $n = 2^{p-1}(2^p - 1)$ with $p$ prime and $2^p - 1$ prime, as required. ∎

**§ Remarks on perfect numbers** - This theorem establishes a bijection between Mersenne primes and even perfect numbers. It is unknown whether there are infinitely many Mersenne primes (hence unknown whether there are infinitely many even perfect numbers). - It is also unknown whether any odd perfect numbers exist; the conjecture "every perfect number is even" remains open.

**§ The Möbius function $\mu$** The Möbius function $\mu : \mathbb{Z}_{>0} \to \{-1, 0, 1\}$ is defined by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & \text{if } p^2 \mid n \text{ for some prime } p, \\ (-1)^r, & \text{if } n = p_1 p_2 \cdots p_r \text{ is a product of } r \text{ distinct primes.} \end{cases}$$

**§ Examples for $\mu$** Since $504 = 2^3 \cdot 3^2 \cdot 7$ contains a squared prime factor, $\mu(504) = 0$. Since $30 = 2 \cdot 3 \cdot 5$ is a product of three distinct primes, $\mu(30) = (-1)^3 = -1$.

**§ Multiplicativity of $\mu$** The Möbius function $\mu(n)$ is multiplicative: if $\gcd(m, n) = 1$ then $\mu(mn) = \mu(m)\mu(n)$.

*Proof.* If either $m$ or $n$ is divisible by a square of a prime then so is $mn$, and both sides are 0. Otherwise write $m = \prod_{i=1}^{r} p_i$ and $n = \prod_{j=1}^{s} q_j$ as products of distinct primes (with the two lists disjoint by coprimality). Then

$$\mu(mn) = (-1)^{r+s} = (-1)^r(-1)^s = \mu(m)\mu(n).$$

∎

**§ Möbius summatory identity (Proposition)** For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d \mid n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

*Proof.* Define $F(n) = \sum_{d|n} \mu(d)$. The function $F$ is multiplicative since it is a divisor-sum of a multiplicative function $\mu$ (Theorems Multiplicativity of $\mu$ and Multiplicative functions and divisor sums). Thus $F$ is determined by its values on prime powers. Clearly $F(1) = 1$. If $p$ is prime and $a \geq 1$ then

$$F(p^a) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^a) = 1 + (-1) + 0 + \cdots + 0 = 0.$$

Hence $F(n) = 0$ for every $n > 1$ (by multiplicativity and prime-power vanishing), proving the claim. □

**§ Example: $n = 12$ (check)** For $n = 12$ the divisors are $1, 2, 3, 4, 6, 12$ and

$$\mu(1) + \mu(2) + \mu(3) + \mu(4) + \mu(6) + \mu(12) = 1 + (-1) + (-1) + 0 + 1 + 0 = 0,$$

consistent with Proposition Möbius summatory identity (Proposition).

## 0.11 L11: 3.6 Möbius inversion, 4.1 Quadratic residues

**§ Möbius Inversion** Let $f$ and $g$ be arithmetic functions (functions $\mathbb{Z}_{>0} \to \mathbb{C}$). Then

$$f(n) = \sum_{d|n} g(d) \quad \text{for all } n \geq 1$$

if and only if

$$g(n) = \sum_{d|n} \mu(d)\, f\left(\frac{n}{d}\right) = \sum_{d|n} \mu\left(\frac{n}{d}\right) f(d) \quad \text{for all } n \geq 1,$$

where $\mu$ denotes the Möbius function.

*Proof.* We prove both directions.

*("⇒")* Assume $f(n) = \sum_{d|n} g(d)$ for every $n \geq 1$. Then for each fixed $n$,

$$\sum_{d|n} \mu(d)\, f\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) \sum_{c|\frac{n}{d}} g(c).$$

The double sum can be rearranged by letting $c$ run over divisors of $n$ and collecting terms in $d$ with $d \mid n/c$:

$$\sum_{c|n} g(c) \sum_{d|n/c} \mu(d).$$

By the well-known identity (see Proposition Möbius summatory identity (Proposition)) the inner sum $\sum_{d|m} \mu(d)$ equals 1 if $m = 1$ and 0 otherwise. Hence the inner sum is 1 exactly when $n/c = 1$, i.e. $c = n$, and is 0 for all other $c$. Thus the whole expression reduces to $g(n)$, proving the claimed formula for $g(n)$.

*("⇐")* Conversely, assume $g(n) = \sum_{d|n} \mu(n/d)\, f(d)$ for all $n$. Then

$$\sum_{d|n} g(d) = \sum_{d|n} \sum_{c|d} \mu(d/c)\, f(c).$$

Rearrange by summing first over $c \mid n$ and then over $d$ with $c \mid d \mid n$:

$$\sum_{c|n} f(c) \sum_{\substack{d \\ c|d,\ d|n}} \mu(d/c) = \sum_{c|n} f(c) \sum_{m|(n/c)} \mu(m).$$

Again, by Proposition Möbius summatory identity (Proposition) the inner sum is 1 if $n/c = 1$ (i.e. $c = n$) and 0 otherwise. Therefore the sum reduces to $f(n)$, proving that $f(n) = \sum_{d|n} g(d)$ for every $n$. □

**§ Remark** The Möbius inversion formula is exactly the statement that $\mu$ is the convolutional inverse of the constant-1 function with respect to Dirichlet convolution. Equivalently, if $f = g * 1$ (Dirichlet convolution), then $g = f * \mu$. (See Homework 4 for exercises on this point.)

**Example.** Let $f(n) = n$ for all $n \geq 1$. By the well-known identity

$$n = \sum_{d|n} \varphi(d),$$

we may take $g(d) = \varphi(d)$. Applying Möbius inversion yields

$$\varphi(n) = \sum_{d|n} \mu(d)\, \frac{n}{d} = \sum_{d|n} \mu\left(\frac{n}{d}\right) d.$$

**Example.** Let $\tau(n)$ denote the number of positive divisors of $n$ (in your notes this was written as $v(n)$; I replaced it by the standard notation $\tau(n)$ and noted this change above). Then

$$\tau(n) = \sum_{d|n} 1,$$

so $g(d) = 1$. By Möbius inversion we obtain the identity

$$1 = \sum_{d|n} \mu(d)\, \tau\Big(\frac{n}{d}\Big) = \sum_{d|n} \mu\Big(\frac{n}{d}\Big)\, \tau(d).$$

**Example.** Let $\sigma(n) = \sum_{d|n} d$ be the sum-of-divisors function. Then $g(d) = d$ and Möbius inversion gives

$$n = \sum_{d|n} \mu(d)\, \sigma\Big(\frac{n}{d}\Big) = \sum_{d|n} \mu\Big(\frac{n}{d}\Big)\, \sigma(d).$$

# Quadratic residues

In the previous sections we studied linear congruences of the form $ax \equiv b \pmod m$. We now turn our attention to *quadratic* congruences; classically one studies congruences of the form

$$x^2 \equiv a \pmod p,$$

especially in the case where $p$ is an odd prime. A central tool is the Legendre symbol $(\frac{a}{p})$ (introduced later), which encodes whether $a$ is a quadratic residue modulo $p$.

**§ Quadratic residue modulo $m$** Let $a, m \in \mathbb{Z}$ with $m > 0$ and $\gcd(a, m) = 1$. We say that $a$ is a *quadratic residue modulo $m$* if the congruence

$$x^2 \equiv a \pmod m$$

has a solution $x \in \mathbb{Z}$. Otherwise $a$ is a *quadratic non-residue modulo $m$*.

**Example (quadratic residues modulo 11).** Compute $x^2 \pmod{11}$ for $x = 1, 2, \ldots, 10$:

$$\begin{array}{l} 1^2 \equiv 1,\ 2^2 \equiv 4,\ 3^2 \equiv 9, \\ 4^2 \equiv 5,\ 5^2 \equiv 3,\ 6^2 \equiv 3, \\ 7^2 \equiv 5,\ 8^2 \equiv 9,\ 9^2 \equiv 4, \\ 10^2 \equiv 1 \end{array} \pmod{11}$$

Thus the distinct quadratic residues modulo 11 (among the reduced residue classes) are

$$1, 3, 4, 5, 9,$$

and the quadratic non-residues (modulo 11) are

$$2, 6, 7, 8, 10.$$

**§ Proposition** Let $p$ be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then the congruence

$$x^2 \equiv a \pmod p$$

has either no solution or exactly two incongruent solutions modulo $p$.

*Proof.* Suppose the congruence has a solution $x_0$. Then $(-x_0)^2 \equiv x_0^2 \equiv a$, so $-x_0$ is also a solution. We claim $x_0 \not\equiv -x_0 \pmod p$. Indeed, if $x_0 \equiv -x_0 \pmod p$ then $2x_0 \equiv 0 \pmod p$. Since $p$ is odd, $\gcd(2, p) = 1$, so this implies $x_0 \equiv 0 \pmod p$, and hence $a \equiv x_0^2 \equiv 0 \pmod p$, contradicting the assumption $p \nmid a$. Thus $x_0$ and $-x_0$ are distinct modulo $p$.

Now suppose $x_0$ and $x_1$ are any two solutions. Then $x_0^2 \equiv x_1^2 \pmod{p}$, so $p \mid (x_0^2 - x_1^2) = (x_0 - x_1)(x_0 + x_1)$. Since $p$ is prime, we must have $p \mid (x_0 - x_1)$ or $p \mid (x_0 + x_1)$. Therefore $x_0 \equiv x_1 \pmod{p}$ or $x_0 \equiv -x_1 \pmod{p}$. This shows there are at most two incongruent solutions. Combined with the earlier observation that if one solution exists then a second distinct one does too, we conclude there are exactly two incongruent solutions in the solvable case. □

**§ Corollary** Let $p$ be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. If $x^2 \equiv a \pmod{p}$ is solvable and $x_0$ is one solution, then the two incongruent solutions modulo $p$ are $x_0$ and $-x_0 \equiv p - x_0$.

**§ Proposition** Let $p$ be an odd prime. There are exactly $\dfrac{p-1}{2}$ distinct (incongruent) quadratic residues modulo $p$ among the nonzero residue classes, and exactly $\dfrac{p-1}{2}$ quadratic non-residues among the nonzero classes.

*Proof.* Consider the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of size $p-1$. For each $x \in (\mathbb{Z}/p\mathbb{Z})^\times$, the square $x^2$ is a nonzero quadratic residue. Note that $x^2 = (-x)^2$, so each nonzero quadratic residue arises from exactly two distinct elements of $(\mathbb{Z}/p\mathbb{Z})^\times$, namely $x$ and $-x$. Therefore the set of distinct nonzero squares has size $(p-1)/2$. Since the set of nonzero residue classes has size $p-1$, the remaining $(p-1)/2$ classes are quadratic non-residues. This completes the proof. □

## 0.12 L12: 4.2 Legendre symbol

# L12: The Legendre Symbol

**§ Definition 4.1 (Legendre symbol)** Let $p$ be an odd prime and let $a \in \mathbb{Z}$. The *Legendre symbol* $(\frac{a}{p})$ is defined by

$$\left(\frac{a}{p}\right) := \begin{cases} 0, & \text{if } p \mid a, \\ 1, & \text{if } p \nmid a \text{ and } a \text{ is a quadratic residue modulo } p, \\ -1, & \text{if } p \nmid a \text{ and } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

**§ Examples** Working modulo 11, the nonzero quadratic residues are the squares

$$1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 9, \quad 4^2 \equiv 5, \quad 5^2 \equiv 3 \pmod{11},$$

so the quadratic residues modulo 11 are

$$1,\ 3,\ 4,\ 5,\ 9.$$

Hence

$$\left(\frac{1}{11}\right) = \left(\frac{3}{11}\right) = \left(\frac{4}{11}\right) = \left(\frac{5}{11}\right) = \left(\frac{9}{11}\right) = 1,$$

and the remaining nonzero residues

$$2,\ 6,\ 7,\ 8,\ 10$$

are quadratic non-residues modulo 11, so their Legendre symbols equal $-1$:

$$\left(\frac{2}{11}\right) = \left(\frac{6}{11}\right) = \left(\frac{7}{11}\right) = \left(\frac{8}{11}\right) = \left(\frac{10}{11}\right) = -1.$$

**§ Discussion**

- The Legendre symbol $\left(\frac{a}{p}\right)$ tells whether the congruence $x^2 \equiv a \pmod{p}$ is solvable (value 1), not solvable (value $-1$), or whether $p \mid a$ (value 0).
- It does not provide the actual solutions of the congruence; if the symbol equals 1, one still must find the square roots mod $p$ by other methods.

**§ Example 4.2** Find $(\frac{3}{7})$. By direct computation of squares modulo 7,

$$1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 2 \pmod{7},$$

so the nonzero quadratic residues modulo 7 are $\{1, 2, 4\}$ (there are $(7-1)/2 = 3$ of them). Therefore 3 is not a quadratic residue modulo 7, and

$$\left(\frac{3}{7}\right) = -1.$$

**§ Theorem 4.4 (Euler's Criterion)** Let $p$ be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p},$$

and, since both sides take values in $\{\pm 1\}$, we may (and usually do) interpret this congruence as the equality

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \pmod{p} \in \{1, -1\}.$$

*Proof.* First suppose $\left(\frac{a}{p}\right) = 1$. Then there exists $x_0$ with $x_0^2 \equiv a \pmod{p}$, so

$$a^{\frac{p-1}{2}} \equiv (x_0^2)^{\frac{p-1}{2}} \equiv x_0^{p-1} \equiv 1 \pmod{p}$$

by Fermat's little theorem. Hence $a^{(p-1)/2} \equiv 1 \pmod{p}$, which matches $\left(\frac{a}{p}\right) = 1$.

Now suppose $\left(\frac{a}{p}\right) = -1$. Consider the set $\{1, 2, \dots, p-1\}$. For each $i \in \{1, \dots, p-1\}$ there is a unique $j \in \{1, \dots, p-1\}$ with $ij \equiv a \pmod{p}$ (since multiplication by $i$ is a permutation of the nonzero residues). If $i = j$ for some such pair then $i^2 \equiv a \pmod{p}$, contradicting that $a$ is a non-residue; hence the elements $1, \dots, p-1$ can be paired into $(p-1)/2$ disjoint pairs whose products are congruent to $a$. Therefore

$$(p-1)! \equiv a^{\frac{p-1}{2}} \pmod{p}.$$

By Wilson's theorem $(p-1)! \equiv -1 \pmod{p}$, so $a^{(p-1)/2} \equiv -1 \pmod{p}$, which matches $\left(\frac{a}{p}\right) = -1$.

Combining both cases proves the statement. □

**§ Example 4.3 (Using Euler's Criterion)** Compute $\left(\frac{3}{7}\right)$ using Euler's criterion:

$$3^{\frac{7-1}{2}} = 3^3 = 27 \equiv 27 - 21 = 6 \equiv -1 \pmod{7},$$

so $\left(\frac{3}{7}\right) = -1$.

**§ Proposition 4.5** Let $p$ be an odd prime and let $a, b \in \mathbb{Z}$ with $p \nmid a$ and $p \nmid b$. Then

1. $\left(\frac{a^2}{p}\right) = 1.$
2. If $a \equiv b \pmod{p}$ then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$.
3. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$ (multiplicativity).

*Proof.* (a) If $p \nmid a$ then $a$ is a solution of $x^2 \equiv a^2 \pmod{p}$, so $(a^2/p) = 1$.

(b) If $a \equiv b \pmod{p}$ then the congruences $x^2 \equiv a \pmod{p}$ and $x^2 \equiv b \pmod{p}$ are identical; hence solvability is the same for both, so the Legendre symbols agree.

(c) Using Euler's criterion,

$$\left(\frac{ab}{p}\right) \equiv (ab)^{\frac{p-1}{2}} \equiv a^{\frac{p-1}{2}} b^{\frac{p-1}{2}} \equiv \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \pmod{p}.$$

Both sides are in $\{\pm 1\}$, so the congruence implies equality:

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

□

**§ Example 4.4** Compute $\left(\dfrac{-11}{7}\right)$. We factor

$$\left(\frac{-11}{7}\right) = \left(\frac{-1}{7}\right)\left(\frac{11}{7}\right).$$

Since $11 \equiv 4 \pmod 7$ and $4 = 2^2$ is a square, $(11/7) = (4/7) = 1$. Hence

$$\left(\frac{-11}{7}\right) = \left(\frac{-1}{7}\right).$$

By Theorem Theorem 4.6 (Value of $(-1/p)$) below (or direct computation) $\left(\dfrac{-1}{7}\right) = -1$ because $7 \equiv 3 \pmod 4$. Thus $\left(\dfrac{-11}{7}\right) = -1$.

**§ Further remarks on reduction to prime factors** Let $p$ be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Write the integer $a$ (up to sign) as a product of prime powers

$$a = \pm 2^{\alpha_0} q_1^{\alpha_1} q_2^{\alpha_2} \cdots q_r^{\alpha_r},$$

where the $q_i$ are distinct odd primes different from $p$ and the $\alpha_i \geq 0$ are integers. By multiplicativity (Proposition Proposition 4.5(c)) we have

$$\left(\frac{a}{p}\right) = \left(\frac{\pm 1}{p}\right)\left(\frac{2}{p}\right)^{\alpha_0}\left(\frac{q_1}{p}\right)^{\alpha_1} \cdots \left(\frac{q_r}{p}\right)^{\alpha_r}.$$

Hence it suffices to compute $\left(\frac{\pm 1}{p}\right)$, $\left(\frac{2}{p}\right)$, and $\left(\frac{q}{p}\right)$ for an odd prime $q \neq p$.

**§ Plan / Roadmap**

- $\left(\dfrac{1}{p}\right) = 1$ by Proposition Proposition 4.5(a).
- Next: compute $\left(\dfrac{-1}{p}\right)$ (done below).
- Next lectures: compute $\left(\dfrac{2}{p}\right)$ and the general reciprocity for odd primes $\left(\dfrac{q}{p}\right)$.

**§ Theorem 4.6 (Value of $(-1/p)$)** Let $p$ be an odd prime. Then

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = \begin{cases} 1, & \text{if } p \equiv 1 \pmod 4, \\ -1, & \text{if } p \equiv 3 \pmod 4. \end{cases}$$

*Proof.* Apply Euler's criterion with $a = -1$:

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod p.$$

But the left-hand side is $\pm 1$ and the right-hand side equals the integer $(-1)^{(p-1)/2} \in \{\pm 1\}$, hence equality holds. Writing $p = 4k+1$ or $p = 4k+3$ and evaluating $(-1)^{(p-1)/2}$ gives the two cases above. $\square$

**§ Examples**

- $\left(\frac{-1}{7}\right) = -1$ because $7 \equiv 3 \pmod 4$.
- $\left(\frac{-1}{53}\right) = 1$ because $53 \equiv 1 \pmod 4$.

**§ Lemma 4.7 (Gauss's Lemma)** Let $p$ be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. For $k = 1, 2, \ldots, \frac{p-1}{2}$ let $r_k$ be the least positive residue of $ka$ modulo $p$ (i.e., $r_k \in \{1, \ldots, p-1\}$ and $r_k \equiv ka \pmod p$). Let $n$ be the number of these residues $r_k$ that are greater than $p/2$. Then

$$\left(\frac{a}{p}\right) = (-1)^n.$$

**§ Example 4.5 (Gauss's Lemma)** Use Gauss's Lemma to compute $\left(\frac{6}{11}\right)$.

Consider the least positive residues of

$$6,\ 2 \cdot 6,\ 3 \cdot 6,\ 4 \cdot 6,\ 5 \cdot 6 \quad \text{mod } 11.$$

We have

$$6 \equiv 6, \quad 12 \equiv 1, \quad 18 \equiv 7, \quad 24 \equiv 2, \quad 30 \equiv 8 \pmod{11},$$

so the least positive residues are $\{6, 1, 7, 2, 8\}$. Those greater than $11/2 = 5.5$ are $6, 7, 8$, so $n = 3$. Hence

$$\left(\frac{6}{11}\right) = (-1)^3 = -1.$$

## 0.13 L13: 4.2 Legendre symbol, 4.3 Quadratic reciprocity

# L13: Gauss' Lemma (continued) and Consequences

**§ Proof of Lemma 4.7 (Gauss's Lemma)**

*Proof.* Let $p$ be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. Recall the notation from the lemma: for $k = 1, 2, \dots, \frac{p-1}{2}$ let $r_k$ denote the least positive residue of $ka$ modulo $p$. Partition these residues into two classes:

$$\{r_1, \dots, r_n\} \quad (\text{those } r_k > p/2) \quad \text{and} \quad \{s_1, \dots, s_m\} \quad (\text{those } s_k < p/2),$$

so $n + m = (p-1)/2$. (No least positive residue equals $p/2$ because $p$ is odd; none equals 0 because $p \nmid a$.)

Consider the $(p-1)/2$ integers

$$p - r_1,\ p - r_2,\ \dots,\ p - r_n,\ s_1,\ s_2,\ \dots,\ s_m. \tag{0.1}$$

Each integer in (0.1) lies in the range $1, \dots, (p-1)/2$. We claim that the integers in (0.1) are pairwise distinct modulo $p$, hence they form a permutation of $\{1, 2, \dots, (p-1)/2\}$.

First, no two of the numbers $p - r_i$ can be congruent modulo $p$, since $p - r_i \equiv p - r_j \pmod p$ would imply $r_i \equiv r_j \pmod p$, contradicting the uniqueness of least residues $r_i \neq r_j$ coming from distinct multipliers $k_i \neq k_j$ (with $1 \leq k_i, k_j \leq (p-1)/2$).

Second, no two of the $s_j$ are congruent modulo $p$ for the same reason: they are distinct least residues.

Third, no $p - r_i$ is congruent to any $s_j$ modulo $p$: otherwise $p - r_i \equiv s_j \pmod p$, hence $r_i + s_j \equiv 0 \pmod p$, so $r_i \equiv -s_j \pmod p$. But $r_i$ arises from $k_i a$ and $s_j$ from $k_j a$ for some $k_i \neq k_j$ in $\{1, \dots, (p-1)/2\}$, so this would give

$$k_i a \equiv -k_j a \pmod p \quad \Longrightarrow \quad (k_i + k_j) a \equiv 0 \pmod p.$$

Since $p \nmid a$, this forces $k_i + k_j \equiv 0 \pmod p$. But $1 \leq k_i, k_j \leq (p-1)/2$ implies $1 \leq k_i + k_j \leq p - 1$, so the only way $k_i + k_j \equiv 0 \pmod p$ is $k_i + k_j = p$, which contradicts $k_i, k_j \leq (p-1)/2$. Thus no such congruence can occur.

Therefore the list (0.1) is a permutation of $1, 2, \dots, (p-1)/2$. Taking the product of all entries of (0.1) gives

$$\prod_{i=1}^{n} (p - r_i) \cdot \prod_{j=1}^{m} s_j \equiv \left(\frac{p-1}{2}\right)! \pmod p.$$

Since $p - r_i \equiv -r_i \pmod p$, this becomes

$$(-1)^n \left(\prod_{i=1}^{n} r_i\right) \left(\prod_{j=1}^{m} s_j\right) \equiv \left(\frac{p-1}{2}\right)! \pmod p.$$

By construction the multiset $\{r_1, \dots, r_n, s_1, \dots, s_m\}$ coincides with the least positive residues of

$$a, 2a, \dots, \left(\frac{p-1}{2}\right) a,$$

so

$$\left(\prod_{i=1}^{n} r_i\right) \left(\prod_{j=1}^{m} s_j\right) \equiv a \cdot 2a \cdots \left(\frac{p-1}{2} a\right) = a^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \pmod p.$$

Combining the last two displayed congruences,

$$(-1)^n a^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \equiv \left(\frac{p-1}{2}\right)! \pmod p.$$

Since $\gcd\big((\frac{p-1}{2})!, p\big) = 1$, we may cancel the factorial modulo $p$ to obtain

$$(-1)^n a^{\frac{p-1}{2}} \equiv 1 \pmod{p},$$

so

$$a^{\frac{p-1}{2}} \equiv (-1)^n \pmod{p}.$$

By Euler's criterion (Theorem Theorem 4.4 (Euler's Criterion)) the left-hand side equals $(a/p)$, which takes the value $\pm 1$, giving

$$\left(\frac{a}{p}\right) = (-1)^n.$$

This completes the proof of Gauss's Lemma. □

**§ Theorem 4.8 (Value of $(2/p)$)** Let $p$ be an odd prime. Then

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = \begin{cases} 1, & \text{if } p \equiv 1, 7 \pmod{8}, \\ -1, & \text{if } p \equiv 3, 5 \pmod{8}. \end{cases}$$

*Proof.* Apply Gauss's Lemma (Lemma Lemma 4.7 (Gauss's Lemma)): let $n$ denote the number of least positive residues of

$$2,\ 2 \cdot 2,\ 3 \cdot 2,\ \ldots,\ \left(\frac{p-1}{2}\right) 2$$

that are greater than $p/2$. Then $\left(\frac{2}{p}\right) = (-1)^n$.

For $1 \le k \le (p-1)/2$ the product $k \cdot 2$ satisfies $k \cdot 2 < p/2$ iff $k < p/4$. Hence exactly $\left\lfloor \frac{p}{4} \right\rfloor$ of the values $k \cdot 2$ produce residues $< p/2$, and thus

$$n = \frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor.$$

We need the parity of $n$. We claim

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor \equiv \frac{p^2-1}{8} \pmod{2}.$$

It suffices to check this congruence for the four residue classes of $p$ modulo 8.

**Case** $p \equiv 1 \pmod{8}$: write $p = 8k + 1$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k - \lfloor 2k + \tfrac{1}{4} \rfloor = 4k - 2k = 2k \equiv 0 \pmod{2},$$

and

$$\frac{p^2-1}{8} = \frac{(8k+1)^2 - 1}{8} = 8k^2 + 2k \equiv 0 \pmod{2}.$$

**Case** $p \equiv 3 \pmod{8}$: write $p = 8k + 3$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k + 1 - \lfloor 2k + \tfrac{3}{4} \rfloor = 4k + 1 - (2k + 0) = 2k + 1 \equiv 1 \pmod{2},$$

and

$$\frac{p^2-1}{8} = \frac{(8k+3)^2 - 1}{8} = 8k^2 + 6k + 1 \equiv 1 \pmod{2}.$$

**Case** $p \equiv 5 \pmod{8}$: write $p = 8k + 5$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k + 2 - \lfloor 2k + 1 + \tfrac{1}{4} \rfloor = 4k + 2 - (2k + 1) = 2k + 1 \equiv 1 \pmod{2},$$

and

$$\frac{p^2-1}{8} = \frac{(8k+5)^2-1}{8} = 8k^2+10k+3 \equiv 1 \pmod 2.$$

**Case** $p \equiv 7 \pmod 8$: write $p = 8k+7$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k+3-\lfloor 2k+1+\tfrac{3}{4} \rfloor = 4k+3-(2k+1) = 2k+2 \equiv 0 \pmod 2,$$

and

$$\frac{p^2-1}{8} = \frac{(8k+7)^2-1}{8} = 8k^2+14k+6 \equiv 0 \pmod 2.$$

Thus in all cases the desired parity congruence holds, and therefore

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}},$$

which yields the stated residue-class characterization. □

**§ Examples**

- $\left(\frac{2}{7}\right) = 1$ since $7 \equiv 7 \pmod 8$.
- $\left(\frac{2}{53}\right) = -1$ since $53 \equiv 5 \pmod 8$.

**§ Quadratic Reciprocity — statement** The law of quadratic reciprocity relates $(\frac{q}{p})$ and $(\frac{p}{q})$ for odd primes $p \neq q$, often simplifying computations of Legendre symbols.

**§ Theorem 4.9 (Quadratic Reciprocity)** Let $p$ and $q$ be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right) \quad \text{unless } p \equiv q \equiv 3 \pmod 4,$$

in which case

$$\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right).$$

**§ Remark** If $p \equiv 1 \pmod 4$ or $q \equiv 1 \pmod 4$, then the sign is $+1$ and $(p/q) = (q/p)$. If $p \equiv q \equiv 3 \pmod 4$, the sign is $-1$ and $(p/q) = -(q/p)$.

**§ Example 4.6: Compute** $(7/53)$ Since $53 \equiv 1 \pmod 4$, by Quadratic Reciprocity (Theorem Theorem 4.9 (Quadratic Reciprocity)) we have

$$\left(\frac{7}{53}\right) = \left(\frac{53}{7}\right).$$

Reduce 53 modulo 7: $53 \equiv 4 \pmod 7$. Thus

$$\left(\frac{7}{53}\right) = \left(\frac{4}{7}\right) = 1,$$

because $4 \equiv 2^2$ is a square modulo 7.

**§ Example 4.7: Compute $\left(\frac{-158}{101}\right)$** We compute

$$\left(\frac{-158}{101}\right) = \left(\frac{-1}{101}\right)\left(\frac{158}{101}\right).$$

Since $101 \equiv 1 \pmod 4$, $\left(\frac{-1}{101}\right) = 1$, so

$$\left(\frac{-158}{101}\right) = \left(\frac{158}{101}\right).$$

Reduce $158 \pmod{101}$: $158 \equiv 57 \pmod{101}$. Factor $57 = 3 \cdot 19$, so multiplicativity gives

$$\left(\frac{57}{101}\right) = \left(\frac{3}{101}\right)\left(\frac{19}{101}\right).$$

Compute $\left(\frac{3}{101}\right)$ by reciprocity. Since $(3-1)/2 = 1$ and $(101-1)/2 = 50$, their product is even, so no sign change occurs:

$$\left(\frac{3}{101}\right) = \left(\frac{101}{3}\right) = \left(\frac{2}{3}\right) = -1,$$

because 2 is a nonresidue modulo 3.

Next compute $\left(\frac{19}{101}\right)$. Here

$$\left(\frac{19}{101}\right) = \left(\frac{101}{19}\right)$$

since $((19-1)/2)((101-1)/2) = 9 \cdot 50$ is even. Reduce $101 \pmod{19}$: $101 \equiv 6 \pmod{19}$, hence

$$\left(\frac{19}{101}\right) = \left(\frac{6}{19}\right) = \left(\frac{2}{19}\right)\left(\frac{3}{19}\right).$$

Using Theorem Theorem 4.8 (Value of $(2/p)$), $\left(\frac{2}{19}\right) = -1$ because $19 \equiv 3 \pmod 8$. Also

$$\left(\frac{3}{19}\right) = -\left(\frac{19}{3}\right)$$

by reciprocity (the sign is $-1$ because $(3-1)/2 \cdot (19-1)/2 = 1 \cdot 9$ is odd). Now $19 \equiv 1 \pmod 3$, so $\left(\frac{19}{3}\right) = \left(\frac{1}{3}\right) = 1$, hence $\left(\frac{3}{19}\right) = -1$. Therefore $\left(\frac{6}{19}\right) = (-1) \cdot (-1) = 1$, so $\left(\frac{19}{101}\right) = 1$.

Combining the two factors,

$$\left(\frac{57}{101}\right) = \left(\frac{3}{101}\right)\left(\frac{19}{101}\right) = (-1) \cdot 1 = -1,$$

and hence

$$\left(\frac{-158}{101}\right) = -1.$$

**§ Warning** Do not apply Quadratic Reciprocity to composite moduli. For example, it is incorrect to write

$$\left(\frac{57}{101}\right) = \left(\frac{101}{57}\right)$$

because 57 is not prime. Quadratic Reciprocity applies only to odd prime moduli.

## 0.14 L14: 4.3 Proof quadratic reciprocity and examples

# L14: Proof of the Quadratic Reciprocity Law (Theorem 4.9)

**§ Lemma 4.10** Let $p$ be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$ and $a$ odd. Define

$$N := \sum_{j=1}^{\frac{p-1}{2}} \left\lfloor \frac{ja}{p} \right\rfloor.$$

Then

$$\left(\frac{a}{p}\right) = (-1)^N.$$

Here $\lfloor\cdot\rfloor$ denotes the floor function.

**§ Remark (role of Lemma 4.10)** Lemma Lemma 4.10, like Euler's criterion and Gauss's Lemma, is often more useful theoretically than computationally; nevertheless it provides an alternate viewpoint that is convenient for proving reciprocity.

**§ Example 4.8 (Use Lemma 4.10 to compute** $(7/11)$**)** Compute $(\frac{7}{11})$ using Lemma Lemma 4.10. For $j = 1, \dots, 5$ compute

$$\left\lfloor \frac{7}{11} \right\rfloor = 0, \quad \left\lfloor \frac{14}{11} \right\rfloor = 1, \quad \left\lfloor \frac{21}{11} \right\rfloor = 1, \quad \left\lfloor \frac{28}{11} \right\rfloor = 2, \quad \left\lfloor \frac{35}{11} \right\rfloor = 3,$$

so

$$N = 0 + 1 + 1 + 2 + 3 = 7.$$

Therefore $(\frac{7}{11}) = (-1)^7 = -1$.

**§ Proof of Lemma 4.10**

*Proof.* Let $p$ and $a$ satisfy the hypotheses. For $k = 1, \dots, \frac{p-1}{2}$ write the division identity

$$ka = p \left\lfloor \frac{ka}{p} \right\rfloor + r_k, \tag{0.2}$$

where $r_k$ is the remainder, $0 \le r_k \le p-1$. Since $p \nmid a$, none of the remainders equals 0. Partition the least positive remainders $r_k$ into

$$\{r_1, \dots, r_n\} \quad (\text{those } r_k > p/2) \quad \text{and} \quad \{s_1, \dots, s_m\} \quad (\text{those } s_k < p/2),$$

so $n + m = (p-1)/2$.

Summing (0.2) over $k = 1, \dots, (p-1)/2$ yields

$$\sum_{k=1}^{\frac{p-1}{2}} ka = p \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor + \sum_{k=1}^{\frac{p-1}{2}} r_k. \tag{2}$$

From the Gauss Lemma proof (cf. Lemma Lemma 4.7 (Gauss's Lemma)), the multiset of remainders $\{r_1, \dots, r_n, s_1, \dots, s_m\}$ coincides with the least positive residues of $a, 2a, \dots, \frac{p-1}{2}a$. Moreover, the integers

$$p - r_1, \dots, p - r_n, s_1, \dots, s_m$$

form a permutation of $1, 2, \ldots, \frac{p-1}{2}$. Hence

$$\sum_{j=1}^{\frac{p-1}{2}} j = \sum_{i=1}^{n}(p - r_i) + \sum_{j=1}^{m} s_j = pn - \sum_{i=1}^{n} r_i + \sum_{j=1}^{m} s_j. \tag{3}$$

But $\sum_{k=1}^{(p-1)/2} r_k = \sum_{i=1}^{n} r_i + \sum_{j=1}^{m} s_j$. Subtracting (3) from (2) gives

$$\sum_{k=1}^{\frac{p-1}{2}} ka - \sum_{j=1}^{\frac{p-1}{2}} j = p \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor - pn + 2\sum_{i=1}^{n} r_i. \tag{4}$$

Reduce both sides of (4) modulo 2. Note that $\sum_{k=1}^{(p-1)/2} ka - \sum_{j=1}^{(p-1)/2} j$ is an integer multiple of $(p-1)/2$ and hence even or odd depending on $(p-1)/2$, but the parity analysis below yields

$$0 \equiv p \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor - pn \pmod 2.$$

Since $p$ is odd we may cancel $p$ modulo 2 to obtain

$$0 \equiv \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor - n \pmod 2,$$

so

$$n \equiv \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor \equiv N \pmod 2.$$

By Gauss's Lemma (Lemma Lemma 4.7 (Gauss's Lemma)) we know $(\frac{a}{p}) = (-1)^n$, hence

$$\left(\frac{a}{p}\right) = (-1)^n = (-1)^N,$$

which proves the lemma. □

**§ Proof of Theorem 4.9 (Quadratic Reciprocity)**

*Proof.* We prove the formula

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$$

for distinct odd primes $p$ and $q$. Without loss of generality assume $p > q$.

Consider the rectangle in the first quadrant with corner coordinates $(0,0)$, $(\frac{p-1}{2}, 0)$, $(\frac{p-1}{2}, \frac{q-1}{2})$, $(0, \frac{q-1}{2})$ and count lattice points with integer coordinates $(x, y)$ satisfying $1 \le x \le \frac{p-1}{2}$ and $1 \le y \le \frac{q-1}{2}$. The total number of such lattice points is

$$\frac{p-1}{2} \cdot \frac{q-1}{2}.$$

We now count these lattice points by grouping them according to whether they lie below or above the line $y = \frac{q}{p}x$ (the diagonal $ON$ in the textbook figure).

Let

$$N_1 := \sum_{j=1}^{\frac{q-1}{2}} \left\lfloor \frac{jp}{q} \right\rfloor \qquad \text{and} \qquad N_2 := \sum_{j=1}^{\frac{p-1}{2}} \left\lfloor \frac{jq}{p} \right\rfloor .$$

A standard lattice-point argument (see Strayer's Figure 4.1) shows that $N_1$ counts the integer lattice points strictly below the line in one triangular region and $N_2$ counts the points in the complementary triangular region, and that every lattice point in the rectangle is counted exactly once by $N_1 + N_2$. Therefore

$$N_1 + N_2 = \frac{p-1}{2} \cdot \frac{q-1}{2}. \tag{5}$$

By Lemma Lemma 4.10 (applied separately to the pairs $(p,q)$ and $(q,p)$) we have

$$\left(\frac{p}{q}\right) = (-1)^{N_1}, \qquad \left(\frac{q}{p}\right) = (-1)^{N_2}.$$

Hence

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{N_1+N_2} = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}},$$

where we used (**??**). This is exactly the statement of Quadratic Reciprocity. □

**§ Example 4.9 (Characterize primes $p$ for which 11 is a quadratic residue)** We seek all odd primes $p \neq 11$ such that $(\frac{11}{p}) = 1$.

By Quadratic Reciprocity (Theorem Theorem 4.9 (Quadratic Reciprocity)),

$$\left(\frac{11}{p}\right) = \begin{cases} \left(\frac{p}{11}\right), & \text{if } p \equiv 1 \pmod 4, \\ -\left(\frac{p}{11}\right), & \text{if } p \equiv 3 \pmod 4. \end{cases}$$

The quadratic residues modulo 11 (nonzero squares) are $1, 3, 4, 5, 9$ and non-residues are $2, 6, 7, 8, 10$.

Thus for $p \equiv 1 \pmod 4$ we need $p \pmod{11} \in \{1, 3, 4, 5, 9\}$, and for $p \equiv 3 \pmod 4$ we need $p \pmod{11} \in \{2, 6, 7, 8, 10\}$. Solving these congruence pairs modulo 44 (via the Chinese Remainder Theorem) yields the residue classes modulo 44 of primes $p$ for which $(\frac{11}{p}) = 1$:

$$p \equiv 1, 5, 9, 25, 37 \pmod{44} \quad \text{or} \quad p \equiv 7, 19, 35, 39, 43 \pmod{44}.$$

Equivalently, the ten congruence classes modulo 44 are

$$1, 5, 7, 9, 19, 25, 35, 37, 39, 43 \pmod{44}.$$

**§ Exercise / Further example** Characterize the primes $p$ for which a given integer $s$ is a quadratic residue modulo $p$ (use factorization, multiplicativity and Quadratic Reciprocity).

### 0.15 L15: 5.1 Primitive roots and order of an integer modulo m.

## L15: Primitive Roots

**§ Definition 5.0 (Order of an integer modulo $m$)** Let $m > 0$ be an integer and let $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. The *order* of $a$ modulo $m$, denoted $\operatorname{ord}_m(a)$, is the least positive integer $n$ such that

$$a^n \equiv 1 \pmod{m}.$$

**§ Remark** By Euler's theorem (Theorem Theorem 4.4 (Euler's Criterion)), $a^{\varphi(m)} \equiv 1 \pmod{m}$, so the order $\operatorname{ord}_m(a)$ exists and satisfies

$$\operatorname{ord}_m(a) \mid \varphi(m).$$

Thus the order is always a divisor of $\varphi(m)$.

**§ Examples (orders modulo 7 and 13)**

- Compute $\operatorname{ord}_7(2)$. One checks
$$2^1 \equiv 2,\ 2^2 \equiv 4,\ 2^3 \equiv 8 \equiv 1 \pmod{7},$$
so $\operatorname{ord}_7(2) = 3$.
- Compute $\operatorname{ord}_7(3)$. One checks
$$3^1 \equiv 3,\ 3^2 \equiv 2,\ 3^3 \equiv 6,\ 3^4 \equiv 4,\ 3^5 \equiv 5,\ 3^6 \equiv 1 \pmod{7},$$
so $\operatorname{ord}_7(3) = 6 = \varphi(7)$.
- Compute $\operatorname{ord}_{13}(2)$. By Proposition 0.15 below the order divides $\varphi(13) = 12$. We compute
$$2^1 \equiv 2,\ 2^2 \equiv 4,\ 2^3 \equiv 8,\ 2^4 \equiv 16 \equiv 3,\ 2^6 \equiv (2^3)^2 \equiv 8^2 \equiv 64 \equiv 12 \equiv -1 \pmod{13}.$$
Since $2^6 \equiv -1 \not\equiv 1$, the order is not $1, 2, 3$, or $6$, hence must be $12$. Thus $\operatorname{ord}_{13}(2) = 12$.

**§ Proposition 5.1** Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For a positive integer $n$ we have

$$a^n \equiv 1 \pmod{m} \iff \operatorname{ord}_m(a) \mid n.$$

In particular $\operatorname{ord}_m(a) \mid \varphi(m)$.

*Proof.* *($\Rightarrow$)* Suppose $a^n \equiv 1 \pmod{m}$. Write $n = q \operatorname{ord}_m(a) + r$ with integers $q \geq 0$ and $0 \leq r < \operatorname{ord}_m(a)$ by the Division Algorithm. Then

$$a^n = a^{q \operatorname{ord}_m(a) + r} = \left(a^{\operatorname{ord}_m(a)}\right)^q a^r \equiv a^r \pmod{m}.$$

Since $a^n \equiv 1 \pmod{m}$, it follows that $a^r \equiv 1 \pmod{m}$. By minimality of $\operatorname{ord}_m(a)$ we must have $r = 0$. Hence $n = q \operatorname{ord}_m(a)$ and $\operatorname{ord}_m(a) \mid n$.

*($\Leftarrow$)* If $\operatorname{ord}_m(a) \mid n$ then $n = \operatorname{ord}_m(a)\, b$ for some $b \in \mathbb{Z}_{>0}$, so

$$a^n = \left(a^{\operatorname{ord}_m(a)}\right)^b \equiv 1^b \equiv 1 \pmod{m}.$$

This completes the equivalence. The last statement follows from Euler's theorem $a^{\varphi(m)} \equiv 1 \pmod{m}$, hence $\operatorname{ord}_m(a) \mid \varphi(m)$ by the first part. $\square$

**§ Proposition 5.2** Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For nonnegative integers $i, j$ we have

$$a^i \equiv a^j \pmod{m} \iff i \equiv j \pmod{\operatorname{ord}_m(a)}.$$

*Proof.* Assume without loss of generality $i \geq j$.

*($\Rightarrow$)* If $a^i \equiv a^j \pmod{m}$, then $a^{i-j} \equiv 1 \pmod{m}$. By Proposition 0.15 we have $\operatorname{ord}_m(a) \mid (i-j)$, so $i \equiv j \pmod{\operatorname{ord}_m(a)}$.

*($\Leftarrow$)* If $i \equiv j \pmod{\operatorname{ord}_m(a)}$ then $i - j = t \operatorname{ord}_m(a)$ for some $t$, and hence

$$a^i = a^{j+t\operatorname{ord}_m(a)} = a^j \left(a^{\operatorname{ord}_m(a)}\right)^t \equiv a^j \cdot 1^t \equiv a^j \pmod{m}.$$

This proves the equivalence. □

**§ Definition 5.1 (Primitive root modulo $m$)** Let $m > 0$. An integer $r$ with $\gcd(r, m) = 1$ is called a *primitive root modulo $m$* if

$$\operatorname{ord}_m(r) = \varphi(m).$$

**§ Examples (primitive roots)**

- 3 is a primitive root modulo 7 since $\operatorname{ord}_7(3) = 6 = \varphi(7)$.
- 2 is a primitive root modulo 13 since $\operatorname{ord}_{13}(2) = 12 = \varphi(13)$ (see above).
- 2 is *not* a primitive root modulo 7 because $\operatorname{ord}_7(2) = 3 \neq 6$.

**§ Nonexistence example (modulo 8)** The reduced residue system modulo 8 is $\{1, 3, 5, 7\}$ and $\varphi(8) = 4$. One checks

$$\operatorname{ord}_8(1) = 1, \quad \operatorname{ord}_8(3) = 2, \quad \operatorname{ord}_8(5) = 2, \quad \operatorname{ord}_8(7) = 2,$$

so no element has order 4. Therefore there are no primitive roots modulo 8.

**§ Proposition 5.3** If $r$ is a primitive root modulo $m$, then the set

$$\{r, r^2, r^3, \ldots, r^{\varphi(m)}\}$$

is a reduced residue system modulo $m$ (i.e., a complete set of representatives of the invertible residue classes mod $m$).

*Proof.* Since $\gcd(r, m) = 1$, each $r^k$ is coprime to $m$. There are $\varphi(m)$ elements in the set; to show they are a reduced residue system it suffices to show they are pairwise incongruent modulo $m$. If $r^i \equiv r^j \pmod{m}$ with $1 \leq i < j \leq \varphi(m)$ then Proposition 0.15 implies $i \equiv j \pmod{\operatorname{ord}_m(r)}$. But $\operatorname{ord}_m(r) = \varphi(m)$, so $0 < j - i < \varphi(m)$ cannot be a multiple of $\varphi(m)$; hence no such congruence can occur and the elements are pairwise distinct modulo $m$. Thus the set is a reduced residue system. □

**§ Remark** If a primitive root modulo $m$ exists, it is typically not unique: if $r$ is a primitive root and $\gcd(k, \varphi(m)) = 1$, then $r^k$ is also a primitive root modulo $m$. In a later lecture we will determine precisely how many primitive roots modulo $m$ exist when they exist at all.

**§ Exercises**

1. Show there are no primitive roots modulo 12.

2. Prove that if $r$ is a primitive root modulo $m$, then exactly $\varphi(\varphi(m))$ of the elements $r^k$ $(1 \leq k \leq \varphi(m))$ are primitive roots modulo $m$ (hint: count $k$ with $\gcd(k, \varphi(m)) = 1$).

## 0.16 L16: 5.1 Primitive roots and order of an integer modulo m. & 5.2 Primitive roots for prime moduli.

# L16: The order of an integer (continued)

**§ Proposition 5.4** Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For any positive integer $i$,

$$\mathrm{ord}_m(a^i) = \frac{\mathrm{ord}_m(a)}{\gcd\big(\mathrm{ord}_m(a),\, i\big)}.$$

**§ Proof**

*Proof.* Write $t := \mathrm{ord}_m(a)$ and set $d = \gcd(t, i)$. Then there exist coprime integers $b, c > 0$ with

$$t = db, \qquad i = dc, \qquad \gcd(b, c) = 1.$$

Compute

$$\big(a^i\big)^b = a^{ib} = a^{(dc)(b)} = a^{tc} = \big(a^t\big)^c \equiv 1 \pmod{m},$$

so by Proposition Proposition 5.1 we have $\mathrm{ord}_m(a^i) \mid b$.

Conversely, since $a^t \equiv 1 \pmod{m}$ we have

$$a^{i\,\mathrm{ord}_m(a^i)} \equiv 1 \pmod{m},$$

so by Proposition Proposition 5.1 it follows that $t \mid i\,\mathrm{ord}_m(a^i)$. Substituting $t = db$ and $i = dc$ gives

$$db \mid dc \cdot \mathrm{ord}_m(a^i) \quad \Longrightarrow \quad b \mid c \cdot \mathrm{ord}_m(a^i).$$

Because $\gcd(b, c) = 1$, this implies $b \mid \mathrm{ord}_m(a^i)$. Combining with the previous divisibility $\mathrm{ord}_m(a^i) \mid b$, we get

$$\mathrm{ord}_m(a^i) = b = \frac{t}{d} = \frac{\mathrm{ord}_m(a)}{\gcd(\mathrm{ord}_m(a), i)}.$$

This completes the proof. □

**§ Corollary 5.5** Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For a positive integer $i$,

$$\mathrm{ord}_m(a^i) = \mathrm{ord}_m(a) \quad \Longleftrightarrow \quad \gcd\big(\mathrm{ord}_m(a),\, i\big) = 1.$$

**§ Proof**

*Proof.* Immediate from Proposition Proposition 5.4: $\mathrm{ord}_m(a^i) = \mathrm{ord}_m(a)/\gcd(\mathrm{ord}_m(a), i)$ equals $\mathrm{ord}_m(a)$ iff the denominator is 1. □

**§ Corollary 5.6 (Counting primitive roots)** If a primitive root modulo $m$ exists, then there are exactly $\varphi(\varphi(m))$ incongruent primitive roots modulo $m$.

**§ Proof**

*Proof.* Let $r$ be a primitive root modulo $m$, so $\mathrm{ord}_m(r) = \varphi(m)$. By Proposition Proposition 5.3, the powers

$$r, r^2, \ldots, r^{\varphi(m)}$$

form a reduced-residue system modulo $m$ and hence represent all $\varphi(m)$ residue classes coprime to $m$. By Corollary Corollary 5.5, $r^i$ is a primitive root (i.e. has order $\varphi(m)$) if and only if $\gcd(\varphi(m), i) = 1$. The number of integers $i$ in $\{1, \dots, \varphi(m)\}$ satisfying this is $\varphi(\varphi(m))$. Thus there are exactly $\varphi(\varphi(m))$ incongruent primitive roots modulo $m$. □

**§ Example 5.1** We know 3 is a primitive root modulo 7. Since $\varphi(7) = 6$, Corollary Corollary 5.6 (Counting primitive roots) predicts $\varphi(\varphi(7)) = \varphi(6) = 2$ primitive roots modulo 7. Indeed, $3^1 \equiv 3$ and $3^5 \equiv 5 \pmod 7$ are the two incongruent primitive roots modulo 7.

**§ Remark** Corollary Corollary 5.6 (Counting primitive roots) applies only when at least one primitive root modulo $m$ exists. For example, although $\varphi(\varphi(8)) = \varphi(4) = 2$, there are no primitive roots modulo 8 (see previous lecture), so the corollary does not apply in that case.

## § Section: Primitive roots for prime moduli

**§ Theorem 5.7 (Lagrange's Theorem for polynomials over $\mathbb{F}_p$)** Let $p$ be a prime and let

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

be a polynomial with integer coefficients of degree $n \geq 1$, and assume $p \nmid a_n$. Then the congruence

$$f(x) \equiv 0 \pmod p$$

has at most $n$ incongruent solutions modulo $p$.

**§ Proof**

*Proof.* We work in the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Proceed by induction on $n$. The base case $n = 1$ is elementary: a nonzero linear polynomial has exactly one root in $\mathbb{F}_p$.

Assume the statement holds for degree $\leq k$ and let $f$ have degree $k+1$. If $f$ has no root then the conclusion is trivial. Otherwise let $a \in \mathbb{F}_p$ be a root. Perform polynomial division in $\mathbb{F}_p[x]$:

$$f(x) = (x-a)q(x) + r,$$

with $q(x) \in \mathbb{F}_p[x]$ of degree $k$ and $r \in \mathbb{F}_p$. Evaluating at $x = a$ gives $0 = f(a) = r$, so $f(x) = (x-a)q(x)$ in $\mathbb{F}_p[x]$. Any other root $b \not\equiv a$ must satisfy $q(b) = 0$, and by the induction hypothesis $q$ has at most $k$ roots. Therefore $f$ has at most $k+1$ roots, completing the induction. □

**§ Proposition 5.8** Let $p$ be a prime and let $d > 0$ be a divisor of $p-1$. Then the congruence

$$x^d \equiv 1 \pmod p$$

has exactly $d$ incongruent solutions modulo $p$.

**§ Proof**

*Proof.* Write $p - 1 = de$ for some integer $e \geq 1$. Factor the polynomial identity

$$x^{p-1} - 1 = x^{de} - 1 = (x^d - 1)\big(x^{d(e-1)} + \cdots + x^d + 1\big).$$

By Fermat's little theorem the congruence $x^{p-1} \equiv 1 \pmod p$ holds for each $x \in \{1, 2, \dots, p-1\}$, so there are $p-1$ solutions among the nonzero residues. Each such solution is a solution of either

$$x^d \equiv 1 \pmod p \quad \text{or} \quad x^{d(e-1)} + \cdots + x^d + 1 \equiv 0 \pmod p.$$

The second polynomial has degree $d(e-1) = p-1-d$, hence by Theorem Theorem 5.7 (Lagrange's Theorem for polynomials over $\mathbb{F}_p$) it has at most $p-1-d$ solutions. Thus the congruence $x^d \equiv 1 \pmod{p}$ must have at least

$$(p-1) - (p-1-d) = d$$

solutions. But Theorem Theorem 5.7 (Lagrange's Theorem for polynomials over $\mathbb{F}_p$) also implies $x^d - 1$ (a degree-$d$ polynomial) has at most $d$ roots modulo $p$. Combining the two bounds yields exactly $d$ distinct solutions modulo $p$, as required. □

**§ Example / Exercise**

- Exercise: Assuming 17 has a primitive root, determine how many incongruent primitive roots there are modulo 17. (*Hint:* use Corollary Corollary 5.6 (Counting primitive roots).)
- Do the same for modulus 18 (be careful: check whether primitive roots modulo 18 exist before applying Corollary Corollary 5.6 (Counting primitive roots)).

**§ Further exercises**

- Prove that 3 is a primitive root modulo 43.
- Using the previous part, find all incongruent integers of order 14 modulo 43.

## 0.17 L17: 5.2 Primitive roots for prime moduli & 5.3 The Primitive root Theorem.

# L17: Primitive Roots for Prime Numbers

**§ Theorem 5.9 (Legendre)** Let $p$ be a prime and let $d$ be a positive divisor of $p-1$ (i.e. $d \mid (p-1)$). Then the number of incongruent integers modulo $p$ whose order equals $d$ is $\varphi(d)$.

**§ Proof**

*Proof.* For each positive divisor $d \mid (p-1)$ let $f(d)$ denote the number of integers $a$ with $1 \le a \le p-1$ such that $\mathrm{ord}_p(a) = d$. By Proposition Proposition 5.1 every nonzero residue mod $p$ has order dividing $p-1$, hence

$$\sum_{d \mid (p-1)} f(d) = p - 1.$$

On the other hand, by elementary properties of the totient function,

$$\sum_{d \mid (p-1)} \varphi(d) = p - 1.$$

Thus

$$\sum_{d \mid (p-1)} f(d) = \sum_{d \mid (p-1)} \varphi(d). \tag{0.3}$$

To conclude $f(d) = \varphi(d)$ for every $d \mid (p-1)$ it suffices to show $f(d) \le \varphi(d)$ for each such $d$. If $f(d) = 0$ this is trivial, so assume $f(d) > 0$. Then there exists $a$ with $\mathrm{ord}_p(a) = d$. The elements

$$a, a^2, \ldots, a^d$$

are pairwise incongruent modulo $p$ (if $a^i \equiv a^j$ with $0 \le i < j \le d$ then $a^{j-i} \equiv 1$ with $1 \le j - i < d$, contradicting minimality of $d$). Hence they are $d$ distinct solutions of

$$x^d \equiv 1 \pmod{p}.$$

By Proposition Proposition 5.8 the congruence $x^d \equiv 1 \pmod{p}$ has exactly $d$ incongruent solutions; therefore the full solution set is precisely $\{a, a^2, \ldots, a^d\}$, which forms a cyclic subgroup of order $d$. Among these $d$ solutions exactly $\varphi(d)$ are generators of the cyclic subgroup (i.e. have order $d$) — that is, exactly the integers $a^k$ with $\gcd(k, d) = 1$ have order $d$. Therefore $f(d) = \varphi(d)$, as required. ☐

**§ Corollary 5.10** Let $p$ be a prime. Then the number of incongruent primitive roots modulo $p$ is $\varphi(p-1)$. In particular, primitive roots modulo $p$ exist for every prime $p$.

**§ Proof**

*Proof.* A primitive root modulo $p$ is precisely an integer of order $p-1$. By Theorem Theorem 5.9 (Legendre), the number of such incongruent integers is $\varphi(p-1)$. In particular $\varphi(p-1) > 0$, so primitive roots exist modulo $p$. ☐

**§ Example 5.2 (Orders modulo 7)** Theorem Theorem 5.9 (Legendre) guarantees existence of integers of orders dividing $\varphi(7) = 6$, namely orders $1, 2, 3, 6$. The count and classes are:

| order $d$ | # of residue classes | residue classes (mod 7) |
|---|---|---|
| 1 | $\varphi(1) = 1$ | $\{1\}$ |
| 2 | $\varphi(2) = 1$ | $\{6\}$ |
| 3 | $\varphi(3) = 2$ | $\{2, 4\}$ |
| 6 | $\varphi(6) = 2$ | $\{3, 5\}$ |

**§ Example 5.3 (Find integers of order 6 modulo 19)** By Corollary Corollary 5.10 there are exactly $\varphi(6) = 2$ incongruent integers of order 6 modulo 19. One convenient approach is to find a primitive root modulo 19 and then use Proposition Proposition 5.4.

First show 2 is a primitive root modulo 19. By Proposition Proposition 5.1 it suffices to check $2^d \not\equiv 1 \pmod{19}$ for each proper divisor $d$ of $18 = 2 \cdot 3^2$, i.e. $d \in \{1, 2, 3, 6, 9\}$. Computations give

$$2^1 \equiv 2,\ 2^2 \equiv 4,\ 2^3 \equiv 8,\ 2^6 \equiv 7,\ 2^9 \equiv 18 \pmod{19},$$

none of which are 1, so $\text{ord}_{19}(2) = 18$. Thus 2 is primitive.

By Proposition Proposition 5.4,

$$\text{ord}_{19}(2^i) = \frac{18}{\gcd(18, i)}.$$

We seek $i$ with $\text{ord}_{19}(2^i) = 6$, i.e. $\frac{18}{\gcd(18,i)} = 6$, so $\gcd(18, i) = 3$. For $1 \leq i \leq 18$ the solutions with $\gcd(18, i) = 3$ are $i = 3, 15$. Hence the elements $2^3$ and $2^{15}$ have order 6.

Compute:

$$2^3 \equiv 8 \pmod{19}.$$

Also

$$2^{15} = 2^{12} \cdot 2^3 \equiv (2^6)^2 \cdot 8 \equiv 7^2 \cdot 8 \equiv 49 \cdot 8 \equiv 11 \cdot 8 \equiv 88 \equiv 12 \pmod{19},$$

so the two incongruent integers with order 6 modulo 19 are $\{8, 12\}$.

**§ Remarks and conjectures**

- Table 5.1 (in your text) lists least primitive roots for small primes; 2 frequently appears as the least primitive root for many primes.
- Conjecture: There are infinitely many primes $p$ for which 2 is the least positive primitive root modulo $p$ (empirical pattern; open).
- **Artin's conjecture:** If $r$ is an integer which is not a perfect square and $r \neq -1$, then there are infinitely many primes $p$ for which $r$ is a primitive root modulo $p$. This remains open; however Heath-Brown (1986) proved that at most two integers $r$ can fail Artin's conjecture in a certain strong sense.

**§ Proposition 5.11** For integers $n \geq 3$ there are no primitive roots modulo $2^n$.

**§ Proof**

*Proof.* If $2^n$ had a primitive root then it would be an odd integer of order $\varphi(2^n) = 2^{n-1}$. We show that every odd integer $a$ satisfies

$$a^{2^{n-2}} \equiv 1 \pmod{2^n},$$

which implies the order divides $2^{n-2} < 2^{n-1}$, hence no primitive root exists.

Proof by induction on $n \geq 3$. Base case $n = 3$: any odd $a \equiv 1, 3, 5, 7 \pmod 8$ satisfies $a^2 \equiv 1 \pmod 8$.

Inductive step: suppose $a^{2^{k-2}} \equiv 1 \pmod{2^k}$ for some $k \geq 3$. Then write

$$a^{2^{k-2}} = 1 + 2^k t$$

for some integer $t$. Squaring both sides gives

$$a^{2^{k-1}} = (1 + 2^k t)^2 = 1 + 2^{k+1} t + 2^{2k} t^2 \equiv 1 \pmod{2^{k+1}},$$

since $2^{2k} t^2$ is divisible by $2^{k+1}$ when $k \geq 3$. This completes the induction and the proof. □

**§ Proposition 5.12** Let $m, n > 2$ be coprime integers. Then there are no primitive roots modulo $mn$.

**§ Proof**

*Proof.* Let $a$ be any integer with $\gcd(a, mn) = 1$. Then $\gcd(a, m) = \gcd(a, n) = 1$. Let

$$d := \gcd\big(\varphi(m), \varphi(n)\big), \qquad k := \operatorname{lcm}\big(\varphi(m), \varphi(n)\big).$$

Since $m, n > 2$ we have $\varphi(m)$ and $\varphi(n)$ even, hence $d \geq 2$. Then

$$k = \frac{\varphi(m)\varphi(n)}{d} = \frac{\varphi(mn)}{d} \leq \frac{\varphi(mn)}{2}.$$

By Euler's theorem $a^{\varphi(m)} \equiv 1 \pmod{m}$, so raising both sides to the $\varphi(n)/d$ power yields

$$a^k \equiv 1 \pmod{m}.$$

Similarly $a^k \equiv 1 \pmod{n}$. Since $(m, n) = 1$, it follows that $a^k \equiv 1 \pmod{mn}$. Therefore the order of $a$ modulo $mn$ divides $k \leq \varphi(mn)/2$, so cannot equal $\varphi(mn)$. Hence no primitive root modulo $mn$ exists. □

**§ Corollary 5.13 (Necessary form for moduli with primitive roots)** If an integer $n > 0$ admits a primitive root then $n$ must be of one of the forms

$$1,\ 2,\ 4,\ p^m,\ 2p^m$$

where $p$ is an odd prime and $m \geq 1$.

**§ Remark / Proof sketch** The full proof of this corollary uses Propositions Proposition 5.11 and Proposition 5.12 and further elementary reduction arguments: any modulus $n$ decomposes as $2^e \prod p_i^{e_i}$; one checks which local factors allow maximal cyclic multiplicative groups and eliminates mixed composite factors. The complete characterization (the Primitive Root Theorem) will be given in a later lecture.

**§ Exercises**

1. Construct the order/residue table for $p = 13$ (analogue of Example 0.17).
2. Find all incongruent integers of order 6 modulo 19 (exercise solved in Example 0.17).
3. Prove that 3 is a primitive root modulo 43, and list all incongruent integers having order 14 modulo 43.
4. Prove Corollary Corollary 5.13 (Necessary form for moduli with primitive roots).

## 0.18 L18: 5.3 The Primitive Root Theorem

# L18: The Primitive Root Theorem (continued)

**§ Lemma 5.14** Let $p$ be an odd prime. There exists a primitive root $r$ modulo $p$ such that

$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$

**§ Proof**

*Proof.* Let $r$ be any primitive root modulo $p$. If $r^{p-1} \not\equiv 1 \pmod{p^2}$ we are done. Otherwise assume

$$r^{p-1} \equiv 1 \pmod{p^2}.$$

Set $r' := r + p$. Since $r' \equiv r \pmod{p}$, $r'$ is a primitive root modulo $p$. We will show $r'^{\,p-1} \not\equiv 1 \pmod{p^2}$.

By the binomial theorem,

$$(r+p)^{p-1} = \sum_{k=0}^{p-1} \binom{p-1}{k} r^{\,p-1-k} p^k.$$

All terms with $k \geq 2$ are divisible by $p^2$, so modulo $p^2$ only the $k = 0$ and $k = 1$ terms contribute:

$$(r+p)^{p-1} \equiv r^{p-1} + \binom{p-1}{1} r^{p-2} p \pmod{p^2}.$$

Since $\binom{p-1}{1} = p - 1$, we get

$$(r+p)^{p-1} \equiv r^{p-1} + (p-1)p\, r^{p-2} \equiv r^{p-1} - p\, r^{p-2} \pmod{p^2},$$

because $(p-1)p \equiv -p \pmod{p^2}$. Using the assumption $r^{p-1} \equiv 1 \pmod{p^2}$ this becomes

$$(r+p)^{p-1} \equiv 1 - p\, r^{p-2} \pmod{p^2}.$$

If $(r+p)^{p-1} \equiv 1 \pmod{p^2}$ then $p\, r^{p-2} \equiv 0 \pmod{p^2}$, so $p \mid r^{p-2}$ and hence $p \mid r$, contradicting $\gcd(r, p) = 1$. Therefore $(r+p)^{p-1} \not\equiv 1 \pmod{p^2}$, and the lemma is proved. □

**§ Corollary 5.15** Let $p$ be an odd prime and let $r$ be a primitive root modulo $p$. Then either $r$ or $r + p$ is a primitive root modulo $p^2$.

**§ Proof**

*Proof.* Let $n := \mathrm{ord}_{p^2}(r)$. By Proposition Proposition 5.1, $n \mid \varphi(p^2) = p(p-1)$. Reducing modulo $p$ shows $\mathrm{ord}_p(r) = p - 1 \mid n$. Hence $n$ is either $p - 1$ or $p(p-1)$. If $n = p(p-1)$ then $r$ is a primitive root modulo $p^2$ and we are done. If $n = p - 1$, then by Lemma Lemma 5.14 the element $r' := r + p$ satisfies $r'^{\,p-1} \not\equiv 1 \pmod{p^2}$; repeating the divisibility argument above yields $\mathrm{ord}_{p^2}(r') \in \{p-1, p(p-1)\}$, but it cannot be $p - 1$ by the lemma, so $\mathrm{ord}_{p^2}(r') = p(p-1)$ and $r'$ is a primitive root modulo $p^2$. □

**§ Lemma 5.16** Let $p$ be an odd prime and let $r$ be a primitive root modulo $p$ with

$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$

For every integer $m \geq 2$,

$$r^{p^{m-2}(p-1)} \not\equiv 1 \pmod{p^m}.$$

**§ Proof**

*Proof.* We prove the claim by induction on $m \geq 2$. The base case $m = 2$ is the hypothesis.

Assume the statement holds for $m = k \geq 2$, i.e.

$$r^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}.$$

Thus there exists an integer $a$ with

$$r^{p^{k-2}(p-1)} = 1 + ap^{k-1},$$

and necessarily $p \nmid a$ (otherwise $r^{p^{k-2}(p-1)} \equiv 1 \pmod{p^k}$, contradicting the induction hypothesis).

Raise both sides to the $p$-th power:

$$r^{p^{k-1}(p-1)} = (1 + ap^{k-1})^p = 1 + p(ap^{k-1}) + \binom{p}{2}(ap^{k-1})^2 + \cdots + (ap^{k-1})^p.$$

All terms except the first two are divisible by $p^{k+1}$, hence modulo $p^{k+1}$ we have

$$r^{p^{k-1}(p-1)} \equiv 1 + ap^k \pmod{p^{k+1}}.$$

Since $p \nmid a$, this is not congruent to 1 modulo $p^{k+1}$. Thus the statement holds for $m = k + 1$. By induction it is true for all $m \geq 2$. □

**§ Proposition 5.17** Let $p$ be an odd prime and $m \geq 1$ a positive integer. Then there exists a primitive root modulo $p^m$.

**§ Proof**

*Proof.* The case $m = 1$ is Corollary Corollary 5.10. For $m \geq 2$, pick a primitive root $r$ modulo $p$ satisfying Lemma Lemma 5.14, and hence Lemma Lemma 5.16 (so that

$$r^{p^{m-2}(p-1)} \not\equiv 1 \pmod{p^m} \quad \text{for the given } m).$$

Let $n := \operatorname{ord}_{p^m}(r)$. By Proposition Proposition 5.1, $n \mid \varphi(p^m) = p^{m-1}(p-1)$. Reducing modulo $p$ gives $\operatorname{ord}_p(r) = p-1 \mid n$. Thus $n = p^k(p-1)$ for some integer $0 \leq k \leq m-1$.

If $k \leq m-2$, then

$$r^{p^{m-2}(p-1)} = \left(r^{p^k(p-1)}\right)^{p^{m-2-k}} \equiv 1^{p^{m-2-k}} \equiv 1 \pmod{p^m},$$

contradicting Lemma Lemma 5.16. Therefore $k = m-1$, and so

$$n = p^{m-1}(p-1) = \varphi(p^m),$$

i.e. $r$ is a primitive root modulo $p^m$. □

**§ Corollary 5.18** Let $p$ be an odd prime and $m \geq 1$. There exists a primitive root modulo $2p^m$.

**§ Proof**

*Proof.* By Proposition Proposition 5.17 choose a primitive root $r$ modulo $p^m$. If $r$ is even, replace it by $r + p^m$ (which is still a primitive root mod $p^m$ and is odd); hence we may assume $r$ is odd and therefore $\gcd(r, 2p^m) = 1$.

Let $n := \operatorname{ord}_{2p^m}(r)$. Then $r^n \equiv 1 \pmod{2p^m}$ implies in particular $r^n \equiv 1 \pmod{p^m}$, so $\varphi(p^m) = p^{m-1}(p-1) \mid n$ by Proposition Proposition 5.1. On the other hand $n \mid \varphi(2p^m) = \varphi(2)\varphi(p^m) = \varphi(p^m)$, hence $n = \varphi(p^m) = \varphi(2p^m)$ and $r$ is a primitive root modulo $2p^m$. □

**§ Theorem 5.19 (Primitive Root Theorem)** Let $n \geq 1$ be an integer. There exists a primitive root modulo $n$ if and only if

$$n \in \{1, 2, 4, p^m, 2p^m\},$$

where $p$ is an odd prime and $m \geq 1$.

**§ Proof**

*Proof.* ($\Rightarrow$) The necessity was proved earlier (Corollary Corollary 5.13 (Necessary form for moduli with primitive roots)): if a primitive root modulo $n$ exists then $n$ must be $1, 2, 4, p^m$, or $2p^m$.

($\Leftarrow$) For each value in the list we exhibit existence.

- $n = 1$: trivial (the residue class 0).
- $n = 2$: 1 is a primitive root modulo 2.
- $n = 4$: 3 is a primitive root modulo 4.
- $n = p^m$ with $p$ odd prime: existence follows from Proposition Proposition 5.17.
- $n = 2p^m$ with $p$ odd prime: existence follows from Corollary Corollary 5.18.

This completes the proof. □

**§ Exercises**

- For each $m \geq 1$, find a primitive root modulo $11^m$.
- Do the same for modulus $17^m$.

### 0.19 L19: 5.4 Index arithmetic and n-th power residues

## L19: Index Arithmetic and $n$th-power residues

**§ Setup / notation** Let $m > 0$ be an integer for which a primitive root exists; fix a primitive root $r$ modulo $m$. Then

$$\{r, r^2, \dots, r^{\varphi(m)}\}$$

is a complete reduced residue system modulo $m$.

**§ Definition (Index)** Let $r$ be a primitive root modulo $m$. For $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$ the *index of $a$ with respect to $r$*, denoted $\mathrm{ind}_r(a)$, is the least positive integer $n$ such that

$$r^n \equiv a \pmod{m}.$$

**§ Remark** Since $\{r^1, \dots, r^{\varphi(m)}\}$ is a reduced-residue system, $\mathrm{ind}_r(a)$ exists and satisfies $1 \le \mathrm{ind}_r(a) \le \varphi(m)$.

**§ Example (indices modulo 7)** Take $r = 3$, a primitive root modulo 7. Compute

$$3^1 \equiv 3,\ 3^2 \equiv 2,\ 3^3 \equiv 6,\ 3^4 \equiv 4,\ 3^5 \equiv 5,\ 3^6 \equiv 1 \pmod{7},$$

hence

$$\mathrm{ind}_3(3) = 1,\ \mathrm{ind}_3(2) = 2,\ \mathrm{ind}_3(6) = 3,\ \mathrm{ind}_3(4) = 4,\ \mathrm{ind}_3(5) = 5,\ \mathrm{ind}_3(1) = 6.$$

**§ Proposition 5.20 (Properties of indices)** Let $r$ be a primitive root modulo $m$ and let $a, b \in \mathbb{Z}$ with $\gcd(a, m) = \gcd(b, m) = 1$. Then

1. $\mathrm{ind}_r(1) = \varphi(m)$ (equivalently $\mathrm{ind}_r(1) \equiv 0 \pmod{\varphi(m)}$).
2. $\mathrm{ind}_r(r) = 1$.
3. $\mathrm{ind}_r(ab) \equiv \mathrm{ind}_r(a) + \mathrm{ind}_r(b) \pmod{\varphi(m)}$.
4. For $n \in \mathbb{Z}_{>0}$: $\mathrm{ind}_r(a^n) \equiv n\,\mathrm{ind}_r(a) \pmod{\varphi(m)}$.

**§ Proof**

*Proof.* (a) and (b) are immediate from definitions: $r^{\varphi(m)} \equiv 1 \pmod{m}$ and $r^1 \equiv r \pmod{m}$. (c) If $r^{\mathrm{ind}_r(a)} \equiv a \pmod{m}$ and $r^{\mathrm{ind}_r(b)} \equiv b \pmod{m}$ then multiplying gives

$$r^{\mathrm{ind}_r(a)+\mathrm{ind}_r(b)} \equiv ab \pmod{m}.$$

By definition $r^{\mathrm{ind}_r(ab)} \equiv ab \pmod{m}$, so by Proposition Proposition 5.2 (periodicity mod $\varphi(m)$) we obtain the congruence in (c). (d) follows similarly: $r^{\mathrm{ind}_r(a^n)} \equiv a^n \equiv (r^{\mathrm{ind}_r(a)})^n \equiv r^{n\,\mathrm{ind}_r(a)} \pmod{m}$, hence the stated congruence. □

**§ Solving multiplicative congruences by indices** Suppose $\gcd(a, m) = \gcd(b, m) = 1$ and $n > 0$. Consider the congruence

$$ax^n \equiv b \pmod{m}. \tag{0.4}$$

Taking indices (base $r$) converts (0.4) into the linear congruence

$$\mathrm{ind}_r(a) + n\,\mathrm{ind}_r(x) \equiv \mathrm{ind}_r(b) \pmod{\varphi(m)},$$

or equivalently

$$n \operatorname{ind}_r(x) \equiv \operatorname{ind}_r(b) - \operatorname{ind}_r(a) \pmod{\varphi(m)}. \tag{0.5}$$

Thus solving (0.4) reduces to solving the linear congruence (0.5) for $\operatorname{ind}_r(x)$; standard linear congruence theory applies.

**§ Example (solve $6x^4 \equiv 3 \pmod 7$ via indices)** Using the primitive root $r = 3$ modulo 7, we have $\operatorname{ind}_3(6) = 3$ and $\operatorname{ind}_3(3) = 1$. The congruence becomes

$$4 \operatorname{ind}_3(x) \equiv \operatorname{ind}_3(3) - \operatorname{ind}_3(6) \equiv 1 - 3 \equiv -2 \equiv 4 \pmod 6.$$

So $4 \operatorname{ind}_3(x) \equiv 4 \pmod 6$. Divide by $\gcd(4, 6) = 2$ to obtain

$$2 \operatorname{ind}_3(x) \equiv 2 \pmod 3 \quad \Longrightarrow \quad \operatorname{ind}_3(x) \equiv 1 \pmod 3.$$

Thus $\operatorname{ind}_3(x) \in \{1, 4\} \pmod 6$, giving

$$x \equiv 3^1 \equiv 3 \pmod 7 \quad \text{or} \quad x \equiv 3^4 \equiv 4 \pmod 7.$$

Hence the two incongruent solutions are $x \equiv 3, 4 \pmod 7$.

**§ Definition ( $n$th-power residue )** Let $a, m, n \in \mathbb{Z}$ with $m > 0, n > 0$ and $\gcd(a, m) = 1$. We say that $a$ is an *$n$th-power residue modulo $m$* if the congruence $x^n \equiv a \pmod m$ has a solution.

**§ Examples**

- 6 is a 3rd-power residue modulo 7 because $3^3 \equiv 6 \pmod 7$.
- 3 is a 4th-power residue modulo 13 because $2^4 \equiv 16 \equiv 3 \pmod{13}$.
- 3 is *not* a 4th-power residue modulo 7 (one checks there is no solution $x$ with $x^4 \equiv 3 \pmod 7$).

**§ Theorem 5.21 (Criterion for $n$th-power residues)** Let $m > 0$ have a primitive root $r$. Fix $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$ and $n > 0$. Let $d = \gcd(n, \varphi(m))$. Then $a$ is an $n$th-power residue modulo $m$ iff

$$a^{\varphi(m)/d} \equiv 1 \pmod m.$$

Moreover, if $a$ is an $n$th-power residue then the congruence $x^n \equiv a \pmod m$ has exactly $d$ incongruent solutions modulo $m$.

**§ Proof**

*Proof.* Write the congruence $x^n \equiv a \pmod m$ in index form (base $r$):

$$n \operatorname{ind}_r(x) \equiv \operatorname{ind}_r(a) \pmod{\varphi(m)}.$$

This linear congruence in $\operatorname{ind}_r(x)$ has a solution iff $d = \gcd(n, \varphi(m))$ divides the right-hand side $\operatorname{ind}_r(a)$. By elementary number theory (Theorem on solvability of linear congruences), this divisibility condition is the solvability criterion.

The divisibility condition $d \mid \operatorname{ind}_r(a)$ is equivalent to

$$\frac{\varphi(m)}{d} \operatorname{ind}_r(a) \equiv 0 \pmod{\varphi(m)},$$

which in turn is equivalent to

$$r^{\frac{\varphi(m)}{d} \operatorname{ind}_r(a)} \equiv 1 \pmod m.$$

But $r^{\mathrm{ind}_r(a)} \equiv a$, so the displayed congruence is exactly $a^{\varphi(m)/d} \equiv 1 \pmod{m}$. Thus the solvability criterion is equivalent to the condition stated.

When solvable, the linear congruence $ny \equiv \mathrm{ind}_r(a) \pmod{\varphi(m)}$ has exactly $d$ incongruent solutions $y \pmod{\varphi(m)}$, hence yields exactly $d$ incongruent values of $x = r^y$ modulo $m$. This proves the counting statement. ☐

**§ Corollary 5.22 (Quadratic residues via Euler)** Let $p$ be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. Then $a$ is a quadratic residue modulo $p$ iff

$$a^{(p-1)/2} \equiv 1 \pmod{p}.$$

Moreover, if $a$ is a quadratic residue then $x^2 \equiv a \pmod{p}$ has exactly two incongruent solutions modulo $p$.

**§ Proof**

*Proof.* Apply Theorem Theorem 5.21 (Criterion for $n$th-power residues) with $m = p$ and $n = 2$. Then $d = \gcd(2, p-1) = 2$, and the criterion becomes $a^{(p-1)/2} \equiv 1 \pmod{p}$. When solvable there are $d = 2$ solutions modulo $p$. ☐

**§ Examples and exercises**

- Example: Take $a = 6, m = 7, n = 3$. Here $\varphi(7) = 6$ and $d = \gcd(3, 6) = 3$. Check $6^{6/3} = 6^2 \equiv 36 \equiv 1 \pmod{7}$, so 6 is a 3rd-power residue modulo 7. By Theorem Theorem 5.21 (Criterion for $n$th-power residues) the congruence $x^3 \equiv 6 \pmod{7}$ has exactly 3 incongruent solutions (indeed they are $x \equiv 3, 3 \cdot 3, 3^3$ etc., in the corresponding index classes).
- Exercise: Is 3 a 4th-power residue modulo 13? If so, how many incongruent solutions does $x^4 \equiv 3 \pmod{13}$ have? (*Hint:* $\varphi(13) = 12$, $d = \gcd(4, 12) = 4$; check $3^{12/4} = 3^3 \pmod{13}$.)
- Example: 15th-power residues modulo 9. Since $9 = 3^2$ has a primitive root and $\varphi(9) = 6$, we get $d = \gcd(15, 6) = 3$. The criterion is $a^{6/3} = a^2 \equiv 1 \pmod{9}$, whose coprime solutions are $a \equiv 1, 8 \pmod{9}$. Thus the 15th-power residues mod 9 are $1, 8$. (Equivalently, the 15-th powers coincide with 3rd powers mod 9 because $15 \equiv 3 \pmod{6}$.)